MATH358 - Advanced Calculus

Differentiation, integration in $\mathbb{R}^n$

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§1 Differentiation

We say $\Omega \subset \mathbb{R}^n$ a *domain* if it is open and connected.

↪**Definition 1.1** (Differentiation): Let $f = (f_1, \ldots, f_m)^T : \Omega \to \mathbb{R}^m$, Ω a domain in $\mathbb{R}^n$ and $f_j : \Omega \to \mathbb{R}$. We say f *differentiable* at $x_0 \in \Omega$ if there exists a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{x \to x_0} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0.$$

Remark 1.1: Note that the first norm on $\mathbb{R}^m$, the second on $\mathbb{R}^n$.

Remark 1.2: In terms of ε, δ, the definition says that $\forall\, \varepsilon > 0,\ \exists\, \delta > 0$ such that if $x \in \Omega \cap B(x_0, \delta)$, then $\|f(x) - f(x_0) - L(x - x_0)\| < \varepsilon \|x - x_0\|$.

↪**Theorem 1.1**: L as above is unique if it exists.

Proof. Suppose $L_1 \neq L_2$ both satisfy the definition. Then, for all $\varepsilon > 0$, there exists $\delta > 0$ such that if $0 < \|x - x_0\| < \delta$, then

$$\begin{aligned} \|(L_1 - L_2)(x - x_0)\| &\leq \|f(x) - f(x_0) - L_1(x - x_0)\| + \|f(x) - f(x_0) - L_2(x - x_0)\| \\ &\leq \varepsilon \|x - x_0\|, \end{aligned}$$

by differentiability (and the previous remark). In particular, $\|(L_1 - L_2)u\| < \varepsilon$ for all unit vectors u, which implies $\|(L_1 - L_2)u\| = 0$ and thus $L_1 = L_2$. ■

↪**Definition 1.2**: If f differentiable at x_0, we'll write $Df(x_0) = L$ for the *differential* of f at x_0.

↪**Proposition 1.1**: f differentiable at x_0 implies f continuous at x_0. In fact, f is Lipschitz at x_0.

Proof. Let $\delta > 0$ such that $\|x - x_0\| < \delta$ implies $\|f(x) - f(x_0) - Df(x_0)(x - x_0)\| < \|x - x_0\|$, which implies

$$\|f(x) - f(x_0)\| \leq \|Df(x_0)(x - x_0)\| + \|x - x_0\| \leq (\|L\| + 1)\|x - x_0\|,$$

which readily proves the statement. ■

↪**Proposition 1.2**: f differentiable at a point x_0 iff each of its component functions are differentiable at x_0.

↪**Definition 1.3**: For $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$, define the *partial derivative*

$$\frac{\partial f_j}{\partial x_i}(x_1, \ldots, x_m) := \lim_{h \to 0} \frac{\left[f_j(x_1, \ldots, x_i + h, \ldots, x_m) - f_j(x_1, \ldots, x_i, \ldots, x_m)\right]}{h},$$

if the limit exists.

↪**Proposition 1.3**: Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable at x_0. Then, $\frac{\partial f_j}{\partial x_i}(x_0)$ exists for each $i = 1, \ldots, n$ and $j = 1, \ldots, m$, and

$$L = Df(x_0) = \left(\frac{\partial f_j}{\partial x_i}(x_0) \right)_{\substack{1 \le i \le n \\ 1 \le j \le m}}.$$

We call this matrix the Jacobian or derivative of f at x_0.

Proof. Write $L = \left(a_{ji}\right)$ in the standard basis $e_1, \ldots, e_n$ for $\mathbb{R}^n$. Let $\varepsilon > 0$, fix some i with $1 \le i \le n$, and set $x := x_0 + he_i$, with $|h| < \delta$ sufficiently small. By differentiability,

$$\frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = \left(\sum_{j=1}^{m} \left[\frac{f_j(x) - f_j(x_0)}{h} - a_{ji} \right]^2 \right)^{1/2}.$$

Since the limit as $h \to 0$ of the above ratio must be zero, the limit of each term in the summation as $h \to 0$ must be zero as well (being a sum of nonnegative terms), i.e.

$$\lim_{h \to 0} \frac{f_j(x) - f_j(x_0)}{h} = a_{ji} \quad \forall\, j = 1, \ldots, m.$$

But the limit on the left is just $\frac{\partial f_j}{\partial x_i}(x_0)$, which proves all of the claims in turn. ■

Remark 1.3: This proposition says that f differentiable at x_0 implies $\frac{\partial f_j}{\partial x_i}(x_0)$ exists for all i, j. The converse need not be true. Consider

$$f(x, y) := \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} & (x, y) \neq 0 \\ 0 & \text{else} \end{cases}.$$

⊛ **Example 1.1**: Another counterexample as in the previous remark is the function

$$f(x,y) := \begin{cases} \frac{x^2y}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}.$$

Claim 1: f continuous at $(0,0)$. We have, for $(x,y) \neq (0,0)$,

$$\begin{aligned} |f(x,y) - f(0,0)| &= \left| \frac{x^2y}{x^2+y^2} \right| \\ &= \frac{x^2|y|}{x^2+y^2} \\ &\leq |y| \to 0 \text{ as } (x,y) \to (0,0), \end{aligned}$$

so we have continuity indeed.

Claim 2: $\partial_x f, \partial_y f$ exist at the origin, and are equal to zero. Note that $f(x,0) = 0$ for $x \neq 0$, and $f(0,0) = 0$, so it follows that $\partial_x f(0,0) = 0$. Similarly for $\partial_y f(0,0)$.

Claim 3: f is not differentiable at $(0,0)$. Suppose otherwise. Then, $L = Df(0,0) = (0,0)$, so

$$\begin{aligned} 0 &= \lim_{(x,y)\to(0,0)} \frac{|f(x,y) - f(0,0) - Df(0,0)(x,y)|}{\|(x,y)\|} \\ &= \lim_{(x,y)\to(0,0)} \frac{|f(x,y)|}{\|(x,y)\|} \\ &= \lim_{(x,y)\to(0,0)} \frac{x^2|y|}{(x^2+y^2)\cdot\sqrt{x^2+y^2}} \\ &= \lim_{(x,y)\to(0,0)} \frac{x^2|y|}{(x^2+y^2)^{3/2}}. \end{aligned}$$

Suppose $y = x$ in the final term (i.e., we approach the limit on a diagonal), and $x > 0$, then this ratio simplifies

$$\frac{x^3}{(2x^2)^{3/2}} = \frac{1}{2^{3/2}} \neq 0,$$

so we have a contradiction.

We can get a partial converse, however, if we assume continuity.

↪**Theorem 1.2**: Let $f = (f_1, \ldots, f_m) : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$. Suppose each $\frac{\partial f_j}{\partial x_i}$ is continuous at some $x^0 \in \Omega$. Then, f is differentiable at x^0.

Proof. We use MVT, and suppose $n = 2, m = 1$ for simplicity of notation, so that $f : \Omega \subset \mathbb{R}^2 \to \mathbb{R}$. We write $x = (x_1, x_2) \in \Omega, x^0 = \left(x_1^0, x_2^0\right)$. Let $\varepsilon > 0$. By assumption, there exists a $\delta > 0$ such that

$$\|y - x^0\| < \delta \Rightarrow \left|\frac{\partial f}{\partial x_i}(y) - \frac{\partial f}{\partial x_i}(x^0)\right| \leq \frac{\varepsilon}{2}, \quad i = 1, 2.$$

We write

$$\begin{aligned} f(x) - f(x^0) &= f(x_1, x_2) - f(x_1^0, x_2) + f(x_1^0, x_2) - f(x_1^0, x_2^0) \\ \text{(MVT, coordinate-wise)} \quad &= \frac{\partial f}{\partial x_1}(z_1, x_2)(x_1 - x_1^0) + \frac{\partial f}{\partial x_2}(x_1^0, z_2)(x_2 - x_2^0), \end{aligned}$$

for some z_1 between x_1 and x_1^0 and some z_2 between x_2 and x_2^0. Thus,

$$\begin{aligned} f(x) - f(x^0) - Df(x^0)(x - x^0) &= f(x) - f(x^0) - \left(\partial_{x_1} f(x^0), \partial_{x_2} f(x^0)\right) \cdot (x - x^0) \\ &= \frac{\partial f}{\partial x_1}(z_1, x_2)(x_1 - x_1^0) + \frac{\partial f}{\partial x_2}(x_1^0, z_2)(x_2 - x_2^0) \\ &\quad - \frac{\partial f}{\partial x_1}(x^0)(x_1 - x_1^0) - \frac{\partial f}{\partial x_2}(x^0)(x_2 - x_2^0) \\ &= \left[\partial_{x_1} f(z_1, x_2) - \partial_{x_1} f(x_1^0, x_2^0)\right](x_1 - x_1^0) \\ &\quad + \left[\partial_{x_2}(x_1^0, z_2) - \partial_{x_2} f(x_1^0, x_2^0)\right](x_2 - x_2^0). \end{aligned}$$

By choice of z_1, z_2 and for (x_1, x_2) in $B(x^0, \delta)$, we know $(z_1, x_2) \in B(x^0, \delta)$ and $(x_1^0, z_2) \in B(x^0, \delta)$ as well, so we can appeal to continuity. In addition, its clear that $|x_i - x_i^0| \leq \|x - x^0\|$. Thus, using continuity, we find

$$|f(x) - f(x^0) - Df(x^0)(x - x^0)| \leq \left(\frac{\varepsilon}{2} + \frac{\varepsilon}{2}\right)\|x - x^0\| = \varepsilon\|x - x^0\|,$$

so dividing both sides by $\|x - x^0\|$ immediately gives the result. ■

↪**Definition 1.4**: Suppose $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ has continuous $\frac{\partial f}{\partial x_i}$ at all points in Ω. Then, we say f is *continuously differentiable* (in Ω), and we write $f \in C^1(\Omega)$.

Remark 1.4: Continuity of partial derivatives is sufficient, but not necessary, for differentiability. For instance,

$$f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2 + y^4} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}.$$

On readily computes $\partial_x f(0, 0) = \partial_y f(0, 0) = 0$, but along the parabola $x = t^2, y = t$ $(t \neq 0)$,

$$\partial_x f(t^2, t) = \frac{1}{2},$$

so $\partial_x f$ can't be continuous. However, f is still differentiable at $(0, 0)$: we claim $L = 0$, then

$$\frac{|f(x, y) - f(0, 0) - L(x, y)|}{\|(x, y)\|} = \frac{|f(x, y)|}{(x^2 + y^2)^{\frac{1}{2}}} = \frac{x^2 y^2}{(x^2 + y^4)(x^2 + y^2)^{\frac{1}{2}}} \leq \frac{y^2}{|x^2 + y^2|^{\frac{1}{2}}} \leq |y| \underset{(x,y) \to 0}{\to} 0.$$

↪**Proposition 1.4** (Basic Properties of Differentiation):

1. If $f, g : \Omega \to \mathbb{R}^m$ both differentiable at $x^0 \in \Omega$, then so is $F = f + g$, and
$$D(f+g)(x^0) = Df(x^0) + Dg(x^0).$$
2. If $f, g : \Omega \to \mathbb{R}^m$ both differentiable at $x^0 \in \Omega$, then so is $F = fg : \Omega \to \mathbb{R}$, and
$$DF(x^0) = f(x^0)Dg(x^0) + g(x^0)Df(x^0).$$
3. $f, g : \Omega \to \mathbb{R}$ both differentiable at x^0 with $g(x^0) \neq 0$, then so is $F = \frac{f}{g}$, and
$$DF(x^0) = \frac{DF(x^0)}{g(x^0)} - \frac{f(x^0)Dg(x^0)}{g^2(x^0)}.$$
4. (Chain Rule) Given $f : \Omega \subset \mathbb{R}^n \to \tilde{\Omega} \subset \mathbb{R}^m$ and $g : \tilde{\Omega} \to \mathbb{R}^k$, with f differentiable at x^0 and g differentiable at $y^0 = f(x^0)$, then $H = g \circ f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^k$ is differentiable at x^0, and
$$DH(x^0) = Dg(y^0) \cdot Df(x^0),$$
in which one should read the "$\cdot$" as matrix multiplication.

Proof. 1., 2., 3. left as an exercise. We prove 4., the Chain Rule, for it is realistically the most interesting. Set $L := Dg(y_0) \cdot Df(x_0)$, and we'll write $y = f(x)$ (so in particular $y_0 = f(x_0)$, as in the statement). We need to show
$$\lim_{x \to x_0} \frac{\|H(x) - H(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0.$$
Let us work the numerator:
$$\begin{aligned} H(x) - H(x_0) - L(x - x_0) &= g(y) - g(y_0) - Dg(y_0)Df(x_0)(x - x_0) \\ &= g(y) - g(y_0) - Dg(y_0)(y - y_0) \\ &\quad + Dg(y_0)(y - y_0) - Dg(y_0)Df(x_0)(x - x_0) \\ &= g(y) - g(y_0) - Dg(y_0)(y - y_0) \\ &\quad + Dg(y_0)(f(x) - f(x_0) - Df(x_0)(x - x_0)). \end{aligned}$$
This means
$$\begin{aligned} \|H(x) - H(x_0) - L(x - x_0)\| \leq & \overbrace{\|g(y) - g(y_0) - Dg(y_0)(y - y_0)\|}^{=:(A)} \\ & + \|Dg(y_0)\| \overbrace{\|f(x) - f(x_0) - Df(x_0)(x - x_0)\|}^{=:(B)}. \end{aligned}$$
By differentiability of f at x_0, $(B) \to 0$ as $\|x - x_0\| \to 0$. We also have that, since f differentiable it is Lipschitz continuous, there is some $C > 0$ such that for $\|x - x_0\|$ sufficiently small,
$$(A) = \|y - y_0\| \cdot \frac{(A)}{\|y - y_0\|} \leq C\|x - x_0\| \frac{A}{\|y - y_0\|}.$$

By differentiability of g, the ratio $\frac{\|A\|}{\|y-y_0\|} \to 0$ as $\|y - y_0\| \to 0$. By continuity of f, $\|y - y_0\| = \|f(x) - f(x_0)\|$ will become small as $\|x - x_0\| \to 0$, so that we have in all $\frac{A}{\|x-x_0\|} \to 0$ as $\|x - x_0\| \to 0$. ■

Exercise 1.1: Let f differentiable in $\mathbb{R}^2$ and $g(r,\theta) := (r\cos\theta, r\sin\theta)$ with $(r,\theta) \in (0,\infty) \times [0, 2\pi)$. Let $F(r,\theta) = f(g(r,\theta))$. Compute $\frac{\partial F}{\partial \theta}$ and $\frac{\partial F}{\partial r}$.

§1.1 Aside on Tangent Planes

Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ differentiable on Ω. Then $Df(x) = \left(\frac{\partial f}{\partial x_1}(x), \ldots, \frac{\partial f}{\partial x_2}(x)\right) =: \nabla f(x)$, called the *gradient* of f. Let $S := \{(x,z) \in \Omega \times \mathbb{R} : z = f(x)\}$ be the *graph* of f. Then, for $x^0 \in \mathbb{R}$,

$$T_{x_0}S = \left\{(x,z) \in \mathbb{R}^n \times \mathbb{R} : z = f(x^0) + \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}(x^0)\left(x_j - x_j^0\right)\right\}$$

is the *tangent plane* to S at x^0.

To see this, let $v \in \mathbb{R}^n$ be a unit vector and $x \in \Omega$. Define $g(t) := f(x + tv)$ for $f : \Omega \to \mathbb{R}$ differentiable (for t sufficiently small, $x + tv$ remains in Ω by openness). We find

$$g'(t) = \langle \nabla f(x + tv), v\rangle$$

for t sufficiently small.

↪**Proposition 1.5**: Suppose $\nabla f(x) \neq 0$. Then, $\nabla f(x)$ points in the direction of steepest increase of f.

Proof. For v a unit vector, the *directional derivative* in the direction of v is $D_v f(x) = \langle \nabla f(x), v\rangle = \|\nabla f(x)\|\cos(\theta)$ where θ the angle between $\nabla f(x)$ and v. This is maximized when $\theta = 0$, i.e. when $v = \frac{\nabla f(x_0)}{\|\nabla f(x_0)\|}$. ■

We can rewrite the graph S as the level set $\{(x,z) \in \Omega \times \mathbb{R} \mid g(x,z) = 0\}$ where $g(x,z) := z - f(x)$. Heuristically, $\nabla g(x_0, z_0)$ should be *normal* to the surface S at (x_0, z_0) (for steepest increase). As such, we define

$$T_{(x_0,z_0)}S := \{\nabla g(x_0,z_0) \cdot (x - x_0, z - z_0) = 0\}.$$

Note that

$$\nabla g(x_0,z_0) = \left(-\partial_{x_1} f(x_0), \ldots, -\partial_{x_n} f(x_0), 1\right),$$

so that

$$T_{(x_0,z_0)} = \{z - z_0 = \nabla f(x_0) \cdot (x - x_0)\},$$

which gives the definition from above.

§1.2 Clairault's Theorem

Here, the question is, given $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ twice differentiable, when can we exchange order of second-order partial derivatives, i.e. when is

$$\frac{\partial^2 f}{\partial x_j \partial x_i} = \frac{\partial^2 f}{\partial x_i \partial x_j}, \quad \forall\, i,j = 1,\dots,n?$$

We need to establish first a generalization of the mean-value theorem. First, note that if

$$\gamma : (a,b) \to \mathbb{R}^n, \quad g : \Omega \subset \mathbb{R}^n \to \mathbb{R}$$

are two differentiable functions with $\gamma((a,b)) \subset \Omega$, then by the chain rule, if we put $H(t) := g(\gamma(t))$,

$$\frac{\partial H}{\partial t} = Dg(\gamma(t)) \cdot D\gamma(t), \quad D\gamma(t) = (\gamma_1'(t),\dots,\gamma_n'(t)).$$

↪**Theorem 1.3** (Mean-Value Theorem): Let $B \subset \mathbb{R}^n$ be a ball and $f : B \to \mathbb{R}$ be differentiable for all $x \in B$. Then, for any $x, y \in B$, there exists $z \in B$ (in particular, we can find z which lies along the line segment between x and y) such that

$$f(x) - f(y) = Df(z) \cdot (x - y).$$

In particular, $|f(x) - f(y)| \leq \|Df(z)\|\|x - y\|$.

Proof. Let $x, y \in B$ fixed and let $\gamma(t) := tx + (1-t)y$ for $t \in [0,1]$. We see that $\gamma(t) \in B$ for all $t \in [0,1]$, and that $D\gamma(t) = x - y$. Set $F(t) := f(\gamma(t))$ (i.e., we restrict f to its values along the straight line along x and y), noting $F : \mathbb{R} \to \mathbb{R}$. So, by 1-dimensional mean-value theorem, there is some $t^* \in [0,1]$ such that

$$\begin{aligned} f(x) - f(y) = F(1) - F(0) = F'(t^*) &= Df\Big(\underbrace{t^*x + (1-t^*)y}_{=:z \in B}\Big) \cdot D\gamma(t) \\ &= Df(z) \cdot (x - y). \end{aligned}$$

■

Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$ differentiable. Remember that $Df : \Omega \to \mathbb{R}^{m \times n}$.

↪**Definition 1.5**: We say f twice differentiable at x if Df exists locally to x and Df is differentiable at x. We write

$$D^2 f = D(Df),$$

and similarly

$$D^k f := D\big(D^{k-1} f\big)$$

with an analogous definition.

↪**Definition 1.6**: Given $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, we see that $f \in C^k(\Omega)$ for $k \in \mathbb{Z}_+$ if all the partial derivatives to order k exist and are continuous in Ω.

↪**Definition 1.7**: If $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ twice differentiable, the *Hessian matrix* is given by

$$H_f(x) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n} \end{pmatrix}.$$

Exercise 1.2: Let $f(x,y) := \begin{cases} \frac{(xy)(x^2-y^2)}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$ and compute $H_f(x,y)$.

↪**Theorem 1.4** (Clairault): Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be twice differentiable at $x \in \Omega$. Then,

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(x) = \frac{\partial^2 f}{\partial x_j \partial x_i}(x), \quad \forall\, i,j = 1,\dots,n.$$

↪**Corollary 1.1**: If $\frac{\partial^2 f}{\partial x_i \partial x_j}$ are all continuous at $x \in \Omega$ for $i,j = 1,\dots,n$, then $\frac{\partial^2 f}{\partial x_i \partial x_j}(x) = \frac{\partial^2 f}{\partial x_j \partial x_i}(x)$.

Proof. (Of Clairault's) It's enough to consider $n = 2$. Fix $(x,y) \in \Omega$, and note that for $s,t \in \mathbb{R}$ sufficiently small, $(x+s, y+t) \in \Omega$ by openness. Set

$$\begin{aligned} \Delta(s,t) &:= f(x+s,y+t) - f(x,y+t) - f(x+s,y) + f(x,y) \\ &= g_t(x+s) - g_t(x), \quad g_t(u) := f(u,y+t) - f(u,y). \end{aligned}$$

By the mean-value theorem, there is some $\xi_{s,t}$ between x and $x+s$ suh that

$$\Delta(s,t) = \frac{\partial g_t}{\partial x}(\xi_{s,t}) \cdot s = \left[\frac{\partial f}{\partial x}(\xi_{s,t}, y+t) - \frac{\partial f}{\partial x}(\xi_{s,t}, y) \right] s. \quad (\ddagger)$$

By assumption, $\frac{\partial f}{\partial x}$ is differentiable at (x,y), so

$$\frac{\partial f}{\partial x}(z_1,z_2) = \frac{\partial f}{\partial x}(x,y)(z_1 - x) + \frac{\partial^2}{\partial x^2}(x,y)(z_2 - y) + E_1(z_1,z_2), \quad (\dagger)$$

where

$$\frac{|E_1(z_1,z_2)|}{\sqrt{(z_1-x)^2 + (z_2-y)^2}} \to 0, \quad \text{as } (z_1,z_2) \to (x,y).$$

Evaluating $(\dagger)$ at $(z_1,z_2) = (\xi_{s,t}, y+t)$ and $(\xi_{s,t}, y)$, and plugging into $(\ddagger)$ yields

$$\Delta(s,t) = \left(\frac{\partial^2 f}{\partial y \partial x}(x,y)t + E_1(\xi_{s,t}, y+t) - E_1(\xi_{s,t}, y) \right) s.$$

Let $s = t$ and let $t \to 0$. We claim that

$$\frac{\partial^2 f}{\partial y \partial x}(x,y) = \lim_{s=t\to 0} \frac{\Delta(s,t)}{st} = \frac{\partial^2 f}{\partial x \partial y}(x,y).$$

The first equality is obvious from the assumptions on the error terms. On the other hand, we can switch the order of the middle terms in $\Delta(s,t)$ and write

$$\begin{aligned}\Delta(s,t) &= f(x+s,y+t) - f(x,y+t) - f(x+s,y) + f(x,y)\\ &= h_s(y+t) - h_s(y), \quad h_s(u) := f(x+s,u) - f(x,u).\end{aligned}$$

Repeating the same argument as above with g_t, we get that

$$\Delta(s,t) = \left(\frac{\partial^2}{\partial x \partial y}(x,y)s + E_2(x+s,\eta_{s,t}) - E_2(x,\eta_{s,t})\right)t,$$

where $\eta_{s,t}$ lies between y and $y+t$, and

$$|E_2(x+s,\eta_{s,t})| \leq |s^2+t^2|, \quad |E_2(x,\eta_{s,t})| \leq \sqrt{s^2+t^2}.$$

Setting $s=t$ here, we get

$$\lim_{\substack{s,t\to 0\\ s=t}} \frac{\Delta(s,t)}{st} = \frac{\partial^2}{\partial x \partial y}(x,y).$$

This proves the claim. ■

§1.3 Inverse Function Theorem

↪**Theorem 1.5** (In 1D): If $f:(a,b)\to(c,d)$ is differentiable with $f'(x)>0$, then there exists $g:(c,d)\to(a,b)$ differentiable such that $y=f(x) \Leftrightarrow x=g(y)$ (i.e. $x=g(y)$).

In higher dimensions, we recall some preliminaries before proving.

↪**Theorem 1.6**: Let (X,d) a complete metric space and $f:X\to X$ a contraction mapping, with $d(f(x_2),f(x_1)) \leq \alpha d(x,y)$ for all $x,y\in X$ for some $0<\alpha<1$. Then, there exists a unique $x_0\in X$ such that $f(x_0)=x_0$.

We will write $M_n := \{n\times n \text{ matrices}\} \cong \mathbb{R}^{n^2}$, and $\|A\| := \sqrt{\sum_{i,j=1}^n a_{ij}^2}$ where $A := \left(a_{ij}\right) \in M_n$. We use

$$\mathrm{GL}(n) := \{A\in M_n : \det(A)\neq 0\} = \det^{-1}(\mathbb{R}\setminus\{0\}), \quad \det : M_n \to \mathbb{R}.$$

Remark that since $\mathbb{R}\setminus\{0\}$ is open, and the map det is continuous (it can be written as a polynomial in the entries a_{ij}'s of the matrix A), we know that $\mathrm{GL}(n)$ an open subset of M_n.

Consider the map

$$f:\mathrm{GL}(n)\to\mathrm{GL}(n), \quad f(A) := A^{-1}.$$

↪**Lemma 1.1**: $\mathrm{GL}(n)\subset M_n$ open and $f\in C^k$ for all $k=1,2,\ldots$.

Proof. We already proved the first statement in our remarks above.

Let $A(j|i)$ be $(n-1)\times(n-1)$ matrix with its jth row and ith columns deleted, then recall

$$\operatorname{adj}(A) = \big((-1)^{i+j} \det A(j|i)\big).$$

By Cramer's formula from linear algebra,

$$f(A) = A^{-1} \frac{1}{\det(A)} \operatorname{adj}(A),$$

which is in C^k since $\det(A)$ C^k (being a polynomial in the coefficients of A) and $\det(A) \neq 0$. ■

↪**Theorem 1.7** (Inverse Function Theorem): Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^n$ be C^1. Let $x_0 \in \Omega$ and assume $Df(x_0) \in \mathrm{GL}(n)$. Then, there exist domains U and V of x_0 and $f(x_0)$ resp. such that $f(U) = V$ and $f|_U$ has a C^1 inverse map $f^{-1} : V \to U$. Moreover, for any $y \in V$ and $x = f^{-1}(y)$, $Df^{-1}(y) = [Df(x)]^{-1}$.

Remark 1.5: By the first lemma above, if $f \in C^k$, $k \geq 1$, we get the same regularity for f^{-1}.

Proof. By translation, its enough to assume $x_0 = f(x_0) = y_0 = 0$ and $Df(x_0) = \mathrm{Id}$ by replacing f with $[Df(0)]^{-1}f$, so we have a mapping

$$f : \Omega \to \mathbb{R}^n, \quad f(0) = 0, Df(0) = \mathrm{Id}.$$

Fix $y \in V$ and set

$$g_y(x) := y + x - f(x),$$

remark that

$$g_y(x) = x \Leftrightarrow y = f(x),$$

so it suffices to show g_y as a mapping $\mathbb{R}^n \to \mathbb{R}^n$ is a contraction mapping, and

$$Dg_y(0) = \mathrm{Id} - \mathrm{Id} = 0.$$

If $f \in C^1(U)$, then $g_y \in C^1(U)$ so that $Dg_y \in C^0(U)$ (similar if $f \in C^k \Rightarrow g_y \in C^k$). Since $Dg_0 \in C^0(U)$, there exists some $\delta > 0$ sufficiently small such that $\|Dg_0(x)\| \leq \frac{1}{2}$, for all $x \in B_\delta(0)$. By mean-value theorem, there exists some $z \in B_\delta(0)$ such that

$$\begin{aligned}
\|g_0(x)\| &= \left\| g_0(x) - \underbrace{g_0(0)}_{=0} \right\| \\
&\leq \|Dg_0(z)\| \|x\| \\
&\leq \frac{\|x\|}{2} < \frac{\delta}{2},
\end{aligned}$$

which implies we can view

$$g_0 : B_\delta(0) \to B_{\delta/2}(0).$$

It follows that

$$g_y : B_\delta(0) \to B_\delta(0), \quad \forall\, y \in B_{\delta/2}(0),$$

using the fact $g_y = y + g_0$ and the triangle inequality. By MVT once again for any $x, x' \in B_\delta(0)$, there exists $y \in B_{\delta/2}(0)$ such that

$$\begin{aligned}\left\|g_y(x) - g_y(x')\right\| &= \|g_0(x) - g_0(x')\| \\ &\leq \|Dg_0(y)\|\|x - x'\| \\ &\leq \frac{\|x - x'\|}{2}\end{aligned}$$

hence $g_y : B_\delta \to B_\delta$ is a contraction mapping. By the fixed-point theorem, there exists a unique point $x \in B_\delta(0)$ such that $g_y(x) = x \Leftrightarrow y = f(x)$. That is, there exists an inverse map $f^{-1} : B_{\delta/2}(0) \to B_\delta(0)$. Moreover, for any $x, x' \in B_\delta(0)$,

$$\begin{aligned}\|x - x'\| &\leq \|f(x) - f(x')\| + \|g_0(x) - g_0(x)\| \\ &\leq \|f(x) - f(x')\| + \frac{1}{2}\|x - x'\|,\end{aligned}$$

i.e.

$$\|x - x'\| \leq 2\|f(x) - f(x')\|.$$

From here, we know that for $y, y' \in B_{\delta/2}(0)$,

$$\left\|f^{-1}(y) - f^{-1}(y')\right\| \leq 2\|y - y'\| \Rightarrow f^{-1} \in C^0\big(B_{\delta/2}(0)\big).$$

Next, we need to show that $Df^{-1}(y)$ exists for $y \in B_{\delta/2}(0)$ for small $\delta > 0$. Since $Df(0) \in \mathrm{GL}(n)$, we know $Df(x) \in \mathrm{GL}(n)$ if $x \in B_\delta(0)$ (possible after shrinking $\delta > 0$). Set

$$W := f^{-1}\big(B_{\delta/2}(0)\big),$$

and choose $R > 0$ suff. small so that

$$\overline{B_R(0)} \subset W.$$

Since $[Df]^{-1} \in C^0\big(\overline{B_R}\big)$ and $\overline{B_R}(0)$ is compact,

$$\left\|[Df(x)]^{-1}\right\| \leq K, x \in \overline{B_r(0)}.$$

Then, given $y, y' \in B_{\delta/2}(0)$ and with $x = f^{-1}(y), x' = f^{-1}(y')$, we find

$$\begin{aligned}\frac{\left\|f^{-1}(y) - f^{-1}(y') - [Df(x')]^{-1}(y - y')\right\|}{\|y - y'\|} &= \frac{\left\|x - x' - [Df(x')]^{-1}(f(x) - f(x'))\right\|}{\|f(x) - f(x')\|} \\ &= \frac{\|x - x'\|}{\|f(x) - f(x')\|}\frac{\left\|Df(x')^{-1}(f(x) - f(x') - Df(x')(x - x'))\right\|}{\|x - x'\|} \\ &\leq 2K\frac{\|f(x) - f(x') - Df(x')(x - x')\|}{\|x - x'\|},\end{aligned}$$

which converges to zero by differentiability of f. This proves the claim $Df^{-1}(y) = [Df(x)]^{-1}$ where $y = f(x)$. ■

Remark 1.6: The inverse function theorem is *local*. In general we can't expect to find a single global inverse. For instance, let

$$f(x,y) := (e^y \cos(x), e^y \sin(x)).$$

One easily verifies

$$\det(Df(x,y)) = e^{-y} \neq 0.$$

However,

$$f(x+2k\pi, y) = f(x,y), \forall\, k \in \mathbb{Z},$$

so there is certainly no hope of a global inverse, for f is not even injective.

↪**Theorem 1.8** (Implicit Function Theorem): Let $F : \Omega \subset \mathbb{R}^n_x \times \mathbb{R}^m_y \to \mathbb{R}^m_y$ be a C^k map. Denote $X = (x,y) \in \mathbb{R}^n \times \mathbb{R}^m$, and let $X_0 = (x_0, y_0) \in \Omega$ with $F(X_0) = 0$. Writing $F = (F_1, \ldots, F_m)$, assume that

$$D_y F(X_0) = \begin{pmatrix} \frac{\partial F_1}{\partial y_1} & \cdots & \frac{\partial F_1}{\partial y_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial y_1} & \cdots & \frac{\partial F_m}{\partial y_m} \end{pmatrix}(X_0)$$

is invertible. Then, there exist neighborhoods U and V of $x_0 \in \mathbb{R}^n$ and $y_0 \in \mathbb{R}^m$ resp. and a unique C^k map $f : U \to V$ such that

$$F(x, f(x)) = 0, \quad \forall\, x \in U.$$

In other words, the level set of F is locally to x_0 the graph of some function f of the same regularity as F.

Proof. Define $G : \Omega \to \mathbb{R}^n \times \mathbb{R}^m$ by

$$G(x,y) := (x, F(x,y)).$$

Obviously G is C^k. We can apply the inverse function theorem to G near X_0; indeed,

$$DG(X_0) = \begin{pmatrix} I_{n\times n} & 0 \\ D_x F(X_0) & D_y F(X_0) \end{pmatrix},$$

which means

$$\det DG(X_0) = \det D_y F(X_0) \neq 0,$$

by assumption. Thus there exist neighborhoods W_1, W_2 of X_0, $(x_0, 0)$ respectively (since $(x_0, 0) = G(X_0)$) for which G^{-1} exists (and is C^k) from $W_2 \to W_1$. Then, there are neighborhoods $U \subset \mathbb{R}^n$ of x_0 and $V \subset \mathbb{R}^m$ of y_0 such that $U \times V \subset W_1$; set $Z = G(U \times V)$ (which is also open, with $Z \subset W_2$). Thus we can view

$$G : U \times V \to Z, \quad G^{-1} : Z \to U \times V,$$

which are both C^k maps. Since $G(x,y) = (x, F(x,y))$, we know that $G^{-1}(x,w) = (x, H(x,w))$ for all $(x,w) \in Z$. Here, $H : Z \to V$ is C^k since G is. Thus,

$$(x, F(x, H(x,w))) = G(x, H(x,w)) = (x,w),$$

using the definition of G in the first equality and the inverse fact in the second line. Thus, it follows that

$$F(x, H(x,w)) = w, \quad \forall\, (x,w) \in Z,$$

thus taking $f(x) := H(x,0)$ gives the proof. ■

↪**Corollary 1.2**: Let $F : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be a $C^k(\Omega)$ function. Let $X = (x', y) \in \mathbb{R}^{n-1} \times \mathbb{R}$ and suppose $(x_0', y_0) \in \Omega$ with $\frac{\partial f}{\partial y}(x_0', y_0) \neq 0$. Then, there exist neighborhoods U and V of $x_0' \in \mathbb{R}^{n-1}$ and $y_0 \in \mathbb{R}$ and a unique $C^k(U)$ function $f : U \to V$ such that

$$\{F(x', y) = 0\} = \{y = f(x')\}, \quad (x', y) \in U \times V.$$

↪**Theorem 1.9** (Morse Lemma): Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be a C^k function with $k \geq 3$. Let $0 \in \Omega$ be a critical point, i.e. $\nabla f(0) = 0$. Assume further $f(0) = 0$ and $\nabla^2 f(0)$ is invertible. There exist open sets U, V of $0 \in U \cap V$ and $g \in C^{k-2}(U)$, $g : U \to V$ with $g^{-1} : V \to U$, $g^{-1} \in C^2(V)$, such that

$$f(g(y)) = y_{\ell+1}^2 + \cdots + y_n^2 - \left(y_1^2 + \cdots + y_\ell^2\right),$$

for some $\ell \in \mathbb{Z} \cap [0, n]$.

§1.4 Taylor's Theorem in $\mathbb{R}^n$ and Lagrange Multipliers

Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, $f \in C^{k+1}(\Omega)$. Let $x_0 \in \Omega$ and $|t|$ small. Consider

$$g(t) := f\left(x_0 + t\frac{x - x_0}{\|x - x_0\|}\right), \quad x \neq x_0, \quad g(0) = f(x_0).$$

Since $x_0 \in \Omega$ and Ω open, $x_0 + t\frac{x-x_0}{\|x-x_0\|} \in \Omega$ for t sufficiently small. By Taylor in 1-dimension,

$$g(t) = g(0) + g'(0)t + \frac{g''(0)t^2}{2!} + \cdots + \frac{g^{(k)}(0)t^k}{k!} + R_k(g)(t), \quad \frac{|R_k(g)(t)|}{|t|^k} \leq M\,|t| \text{ as } t \to 0.$$

Remark 1.7: This expansion suggests that one should be able to weaken the C^{k+1} assumption on g if we only require

$$\lim_{t \to 0} \frac{|R_k(g)(t)|}{|t|^k} = 0.$$

Indeed, we only really need $g \in C^{k-1}$ and $D\left(D^{k-1}\right)g(0)$ existing.

To get the Taylor expansion for $f(x)$ around x_0, we set $t = |x - x_0|$ and apply chain rule to $g(t)$. First, we compute $g^{(j)}(0)$; we get

$$g(0) = f(x_0),$$
$$g(\|x - x_0\|) = f(x).$$

By chain rule,

$$g'(0) = \sum_{j=1}^{n} \frac{\partial f}{\partial x_i}(x_0) \frac{x_j - x_j^0}{\|x - x_0\|} = \nabla f(x_0)^T \left(\frac{x - x_0}{\|x - x_0\|} \right).$$

Similarly,

$$\begin{aligned}
g''(t) &= \frac{\mathrm{d}}{\mathrm{d}t} g'(t) \\
&= \frac{\mathrm{d}}{\mathrm{d}t} \left(\nabla f \left(x_0 + t \frac{x - x_0}{\|x - x_0\|} \right)^T \frac{x - x_0}{\|x - x_0\|} \right) \\
&= \left(D^2 f \left(x_0 + t \frac{x - x_0}{\|x - x_0\|} \right) \frac{x - x_0}{\|x - x_0\|} \right) \cdot \frac{x - x_0}{\|x - x_0\|} \\
&= \left(\frac{x - x_0}{\|x - x_0\|} \right)^T D^2 f \left(x_0 + t \frac{x - x_0}{\|x - x_0\|} \right) \frac{x - x_0}{\|x - x_0\|} \\
&= \sum_{i,j=1}^{n} \frac{\partial^2 f}{\partial x_i \partial x_j} \left(x_0 + t \frac{x - x_0}{\|x - x_0\|} \right) \frac{\left(x_i - x_i^0\right)\left(x_j - x_j^0\right)}{\|x - x_0\|^2},
\end{aligned}$$

so that

$$g''(0) = \left[\frac{x - x_0}{\|x - x_0\|} \right]^T D^2 f(x_0) \cdot \left[\frac{x - x_0}{\|x - x_0\|} \right].$$

Similar computation can be used to compute $g^{(\ell)}$ for ℓ up to k. In general, we find that

$$g^{(\ell)}(0) = \sum_{i_1, \ldots, i_\ell = 1}^{n} \frac{\partial^\ell f}{\partial x_{i_1} \cdots \partial x_{i_\ell}}(x_0) \frac{x_{i_1} - x_{i_1}^0}{\|x - x_0\|} \cdots \frac{x_{i_\ell} - x_{i_\ell}^0}{\|x - x_0\|}.$$

Moreover, since $t^\ell = \|x - x_0\|^\ell$, the term in the Taylor expansion of g corresponding to $g^{(\ell)}(0)\frac{t^\ell}{\ell!}$ becomes $\frac{1}{\ell!}$ times the above, with the denominator $\|\ldots\|$ terms cancelling. This leads to the following theorem.

↪**Theorem 1.10** (Taylor in $\mathbb{R}^n$): Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}, f \in C^{k+1}(\Omega)$. Then, for any $x^0 \in \Omega$,

$$f(x) = f(x^0) + \sum_{j=1}^{k} \frac{1}{j!} \left[\sum_{i_1, \ldots, i_j = 1}^{n} \frac{\partial^j f}{\partial x_{i_1} \cdots \partial x_{i_j}} \left(x_{i_1} - x_{i_1}^0\right) \cdots \left(x_{i_\ell} - x_{i_j}^0\right) \right] + R_k(f)(x),$$

where

$$\frac{|R_k(f)(x)|}{\|x - x^0\|^k} \leq M_k \|x - x^0\| \quad \text{as } x \to x^0.$$

Remark 1.8: The k-th order Taylor polynomial of f is the quantity

$$P_k(f)(x) = f(x_0) + \sum_{j=1}^{k} \frac{1}{j!}\left[\sum_{i_1,\dots,i_j=1}^{n} \frac{\partial^j f}{\partial x_{i_1}\cdots\partial x_{i_j}}\left(x_{i_1} - x_{i_1}^0\right)\cdots\left(x_{i_\ell} - x_{i_j}^0\right)\right]$$

before the remainder, and is a "good approximation" to $f(x)$ for x near x^0, provided

$$\lim_{x\to x^0} \frac{|R_k(f)(x)|}{\|x - x^0\|^k} = 0.$$

If we just make this assumption on the remainder term, we get the following more general result with weaker assumptions:

↪**Theorem 1.11**: Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be $C^{k-1}(\Omega)$ and assume $D^{k-1}f$ is differentiable at $x^0 \in \Omega$. Then the kth order Taylor expansion for f about x^0 from the previous theorem still holds, but with now

$$\lim_{x\to x^0} \frac{|R_k(f)(x)|}{\|x - x^0\|^k} = 0.$$

Remark 1.9: Of course, we lose the rate of decay of the remainder, but we gain fewer assumptions needed on f.

⊛ **Example 1.2**: Let $f(x,y) = e^x + \sin(xy)$. Then

$$f(0,0) = 1,$$

$$f_x(0,0) = 1, \quad f_y(0,0) = 0,$$

$$f_{xx}(0,0) = 1, \quad f_{xy}(0,0) = 1$$

$$f_{yx}(0,0) = 1, \quad f_{yy}(0,0) = 0.$$

Thus,

$$f(x,y) = 1 + x + \frac{x^2}{2} + xy + R_2(f)(x,y),$$

with

$$\left|\frac{R_s(f)(x,y)}{x^2+y^2}\right| \le M\sqrt{x^2+y^2}.$$

§1.5 Lagrange Multipliers

The basic problem is, given $f, g : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ both $C^1(\Omega)$, putting $\Sigma := \{x \in \Omega : g(x) = 0\}$, to find

$$\max f|_\Sigma, \quad \min f|_\Sigma.$$

We call $g : \Omega \to \mathbb{R}$ the "constraint function" (i.e., we are doing constrained optimization of f subject to g).

↪**Theorem 1.12**: Let f, g as above, and assume $Dg(x^0) \neq 0$ for all $x^0 \in \Sigma$. Then, if $f|_\Sigma$ has a max or min at $x_0 \in \Sigma$, then there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(x_0) = \lambda \nabla g(x_0).$$

Proof. Let $x^0 \in \Sigma$. Since by assumption $\nabla g(x_0) \neq 0$, then we can assume (by possibly relabelling the coordinates in $\mathbb{R}^n$) that $\frac{\partial g}{\partial x_n}(x^0) \neq 0$. We can write

$$\Sigma = \{g(x', x_n) = 0\},$$

with $x' = (x_1, \ldots, x_{n-1})$. By IFT, Σ then is of the form

$$x_n = h(x'),$$

for $x' \in \mathbb{R}^{n-1}$ for $(x', x_n) \in \Omega$ near $x^0 \in \Sigma$, where $h \in C^1$ locally to $x_0' := \left(x_1^0, \ldots, x_{n-1}^0\right)$. Now, we set

$$F = f|_\Sigma,$$

so that

$$F(x') = f(x', h(x'))$$

locally to x_0'. Since x' a local max/min of F, we know that $\nabla F(x') = 0$. Applying chain rule (valid since $h \in C^1$) above, we also know that

$$0 = \nabla_{x'} F(x_0') = \nabla_{x'} f(x_0', h(x_0')) = \left(\frac{\partial f}{\partial x_i}(x_0) + \frac{\partial h}{\partial x_i}(x_0') \frac{\partial f}{\partial x_n}(x_0) \right)_{i=1}^{n-1}. \quad (\dagger)$$

We also know that

$$0 = G(x') := g(x', h(x'))$$

for all x' local to x_0', and hence $\nabla G(x') = 0$ local to x_0' as well, i.e.

$$0 = \nabla_{x'} G(x_0') = \left(\frac{\partial g}{\partial x_i}(x_0) + \frac{\partial h}{\partial x_i'}(x_0') \frac{\partial g}{\partial x_n}(x_0) \right)_{i=1}^{n-1}. \quad (\ddagger)$$

Since $\frac{\partial g}{\partial x_n}(x_0) \neq 0$ by assumption, we use $(\ddagger)$ to isolate

$$\frac{\partial h}{\partial x_i'}(x_0') = -\left[\frac{\partial g}{\partial x_n}(x_0) \right]^{-1} \frac{\partial g}{\partial x_i}(x_0), \quad i = 1, \ldots, n-1.$$

Plugging these expressions into $(\dagger)$, we obtain

$$\frac{\partial f}{\partial x_i}(x_0) = \left[\frac{\partial g}{\partial x_n}(x_0) \right]^{-1} \frac{\partial f}{\partial x_n}(x_0) \frac{\partial g}{\partial x_i}(x_0), \quad i = 1, \ldots, n-1.$$

This implies

$$\nabla f(x_0) = \left(\frac{\partial f}{\partial x_1}(x_0), \dots, \frac{\partial f}{\partial x_{n-1}}(x_0), \frac{\partial f}{\partial x_n}(x_0)\right) = \underbrace{\left[\frac{\frac{\partial f}{\partial x_n}(x_0)}{\frac{\partial g}{\partial x_n}(x_0)}\right]}_{=\lambda}\left(\frac{\partial g}{\partial x_1}(x_0), \dots, \frac{\partial g}{\partial x_{n-1}}(x_0), \frac{\partial g}{\partial x_n}(x_0)\right) = \lambda \nabla g(x_0),$$

as claimed (remarking that equality for the first $n-1$ entries follows from our work above, and the last n is trivial since then the denominator of λ cancels with the nth entry of ∇g.) ■

↪**Theorem 1.13** (Max/min): Let $\Omega \subset \mathbb{R}^n$ be a domain with bounded closure, with $\Omega \subset \tilde{\Omega}$ and assume $f, g : \tilde{\Omega} \to \mathbb{R}$ are $C^1(\tilde{\Omega})$, where

$$\partial\Omega = \{x \in \tilde{\Omega} \mid g(x) = 0\}$$

(we say g the *defining function* of $\partial\Omega$). Then,

1. if $x^0 \in \partial\Omega$ a local max/min of $f|_{\partial\Omega}$ then $\nabla f(x^0) = \lambda \nabla g(x^0)$ for some $\lambda \in \mathbb{R}$,
2. if $x^0 \in \operatorname{int}(\Omega)$ a local max/min of f in $\operatorname{int}(\Omega)$, then $\nabla f(x^0) = 0$, and
3. the global max (resp. min) of $f|_{\overline{\Omega}}$ is largest (resp. smallest) value of f at all points in 1., 2.

Proof. Apply the previous result with $\Sigma = \partial\Omega$. ■

↪**Theorem 1.14** (Multiple Constraints): Let $f, g_i : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ $i = 1, \dots, k$ are all $C^1(\Omega)$. Let

$$\Gamma_0 := \{x \in \Omega \mid g_i(x) = 0, \quad i = 1, \dots, k\}$$

and assume

$$\{\nabla g_1(x), \dots, \nabla g_k(x)\}$$

are linearly independent for $x \in \Gamma_0$. Then if $f|_{\Gamma_0}$ has max/min at $x^0 \in \Gamma_0$, there exist $\lambda_1, \dots, \lambda_k$ such that

$$\nabla f(x^0) = \sum_{i=1}^{k} \lambda_i \nabla g_i(x^0).$$

⊛ **Example 1.3**: Find the absolute max/min of $f(x_1, \dots, x_n) = x_1 + x_2$ in $D_1(0) = \{|x|^2 \leq 1\}$. We take $g(x) = |x|^2 - 1$. Since $\nabla f(x) = (1, 1, 0, \dots, 0)$ there cannot be any interior critical points. By Lagrange, at $x_0 \in \partial D_1$, a max/min occuring implies

$$\nabla f(x^0) = \lambda \nabla g(x^0) \Leftrightarrow (1, 1, 0, \dots, 0) = 2\lambda\left(x_1^0, \dots, x_n^0\right).$$

Note that λ can't be zero, thus $x_3^0, \dots, x_n^0 = 0$ and $x_1^0 = \frac{1}{2\lambda} = x_2^0$. Since $x_1^0, x_2^0 \in \partial D_1(0)$ as well, we know $\left(x_1^0\right)^2 + \left(x_2^0\right)^2 = 1$. This implies $\lambda = \pm\sqrt{\frac{1}{2}}$ and so $x_1^0 = x_2^0 = \pm\frac{1}{\sqrt{2}}$. We see that

$$f\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, \dots, 0\right) = \sqrt{2}, \quad f\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0, \dots, 0\right) = -\sqrt{2},$$

thus the global max of f is $\sqrt{2}$ and the global min of f is $-\sqrt{2}$.

§2 Riemann Integration

§2.1 Integration over Box Domains in $\mathbb{R}^n$

A *box* $B \subset \mathbb{R}^n$ is a closed and bounded subset of the form

$$B = [a_1, b_1] \times \cdots \times [a_n, b_n].$$

Given a function $f : B \to \mathbb{R}$, we want to define the Riemann integral (we will drop the "Riemann" specification from here on) of f over B, i.e. $\int_B f \, \mathrm{d}V$.

We know that in $n = 1$, for a function $h : [a, b] \to \mathbb{R}$ (assume nonnegative for simplicity), then

$$\int_a^b h(x) \, \mathrm{d}x \overset{\text{if it exists}}{=} \lim_{n \to \infty} \sum_{i=1}^{n} h(x_i) \Delta x_i,$$

where $a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b$ a partition of $[a, b]$ and $\Delta x_i = x_i - x_{i-1}$. With this as "motivation", say, we will define the integral of f over B.

↪**Definition 2.1** (Partition of a box): Let B a box as above. A *partition* of B is an ordered set

$$\mathcal{P} = \{P_i\}_{i=1}^n, \quad P_i = \left\{x_i^0, \ldots, x_i^{m_i}\right\},$$

where each P_i a partition of $[a_i, b_i]$ with $a_i = x_i^0 < x_i^1 < \cdots < x_i^{m_i - 1} < x_i^{m_i} = b_i$.

↪**Definition 2.2** (Refinement): Given two partitions $\mathcal{P} = \{P_i\}$, $\widetilde{\mathcal{P}} = \{\tilde{P}_i\}$, we say that $\widetilde{\mathcal{P}}$ is a *refinement* of $\mathcal{P}$ (and write $\widetilde{\mathcal{P}} \supset \mathcal{P}$) if for each $j = 1, \ldots, n$, $P_j \subset \tilde{P}_j$.

↪**Definition 2.3** (Subbox): Given a partition $\mathcal{P}$ of B, let

$$B_{i_1, i_2, \ldots, i_n} \subseteq B = \left[x_1^{i_1 - 1}, x_1^{i_1}\right] \times \cdots \times \left[x_n^{i_n - 1}, x_n^{i_n}\right]$$

be a *subbox* of B, for some $\{i_1, \ldots, i_n\}$ partition indices for $\mathcal{P}$ (i.e., $1 \leq i_j \leq m_j$ for $j = 1, \ldots, n$). We define then

$$\Delta x_j^{i_j} := x_j^{i_j} - x_j^{i_j - 1}, \quad j = 1, \ldots, n, 1 \leq i_j \leq m_j,$$

for the length of the jth interval in the subbox, and

$$\Delta V_{i_1, \ldots, i_n} := \Delta x_1^{i_1} \cdots \Delta x_n^{i_n}$$

as thus the volume of the subbox $B_{i_1, \ldots, i_n}$.

Heuristically, then, we'd like to define $\int_B f \, \mathrm{d}V$ as

$$\lim_{m_1 \to \infty} \cdots \lim_{m_n \to \infty} \sum_{i_1 = 1}^{m_1} \cdots \sum_{i_n = 1}^{m_n} f\left(x_1^{i_1}, \ldots, x_n^{i_n}\right) \Delta V_{i_1, \ldots, i_n}.$$

The key question, then, is when does the sum on the right-hand side exist?

↪**Definition 2.4** (Lower/Upper Riemann Sum): Let $f : B \to \mathbb{R}$ be bounded and $\mathcal{P}$ a partition of $\mathcal{B}$. Define the *lower Riemann sum* of f over B corresponding to partition $\mathcal{P}$ by

$$L(f, \mathcal{P}) := \sum_{i_1=1}^{m_1} \cdots \sum_{i_n=1}^{m_n} \left(\inf_{x \in B_{i_1,\ldots,i_n}} f(x) \right) \Delta V_{i_1,\ldots,i_n}.$$

Similarly define the *upper Riemann sum* (of f over B corresponding to partition $\mathcal{P}$) by

$$U(f, \mathcal{P}) := \sum_{i_1=1}^{m_1} \cdots \sum_{i_n=1}^{m_n} \left(\sup_{x \in B_{i_1,\ldots,i_n}} f(x) \right) \Delta V_{i_1,\ldots,i_n}.$$

Remark 2.1: This infs/sups are well-defined since f bounded.

↪**Lemma 2.1**: Let $f : B \to \mathbb{R}$ be bounded and $\widetilde{\mathcal{P}} \supset \mathcal{P}$ a refinement. Then,

$$L(f, \mathcal{P}) \leq L\left(f, \widetilde{\mathcal{P}}\right) \leq U\left(f, \widetilde{\mathcal{P}}\right) \leq U(f, \mathcal{P}).$$

Proof. The intermediate inequality is obvious.

Since $\widetilde{\mathcal{P}} \supset \mathcal{P}$, then there exists $N_{i_j} \in \mathbb{Z}^+$ and $i_{1,\ell_1}, \ldots, i_{n,\ell_n}$ (indices of the refined partition) such that

$$B_{i_1,\ldots,i_n} = \bigcup_{\ell_1=1}^{N_{i_1}} \cdots \bigcup_{\ell_n=1}^{N_{i_n}} \tilde{B}_{i_{1,\ell_1},\ldots,i_{n,\ell_n}},$$

with $\tilde{\cdot}$ used to denote objects belonging to the refinement. Thus,

$$\inf_{x \in B_{i_1,\ldots i_n}} f(x) \leq \inf_{x \in \tilde{B}_{i_{1,\ell_1},\ldots i_{n,\ell_n}}} f(x), \quad \sup_{x \in B_{i_1,\ldots i_n}} f(x) \geq \sup_{x \in \tilde{B}_{i_{1,\ell_1},\ldots i_{n,\ell_n}}} f(x).$$

Summing over the $i_1, \ldots, i_n$'s and multiplying by the appropriate volumes gives the result. ■

↪**Definition 2.5**: Let $f : B \to \mathbb{R}$ bounded. Define

$$\mathcal{U}(f) := \inf_{\mathcal{P}} \{U(f, \mathcal{P})\}, \quad \mathcal{L}(f) := \sup_{\mathcal{P}} \{L(f, \mathcal{P})\}.$$

Remark 2.2: If f bounded with $m \leq f \leq M$ over B, then

$$m \operatorname{vol}(B) \leq U(f, \mathcal{P}) \leq M \operatorname{vol}(B)$$

for any partition $\mathcal{P}$, similar for the lower Riemann sum. This implies these two quantities above are well-defined (infs/sups taken over bounded sets). Moreover, we take inf/sup respectively for the upper/lower sums because of the respective monotonicity properties we proved in the previous lemma.

↪**Definition 2.6** (Riemann integrable): Let $f : B \to \mathbb{R}$ be bounded. Then, we say that f is *Riemann integrable* (RI) over B if $\mathcal{L}(f) = \mathcal{U}(f)$, in which case we define

$$\int_B f \, \mathrm{d}V := \mathcal{L}(f) = \mathcal{U}(f).$$

↪**Lemma 2.2**: Let $f : B \to \mathbb{R}$ be bounded then f is RI over B iff $\forall\, \varepsilon > 0$, there exists a partition $\mathcal{P}$ of B such that

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) \leq \varepsilon.$$

Proof. ($\Leftarrow$) We know $U(f, \mathcal{P}) \geq \mathcal{U}(f)$ and $L(f, \mathcal{P}) \leq \mathcal{L}(f)$, so this is immediate since then we conclude that for all $\varepsilon > 0$,

$$\mathcal{U}(f) \leq U(f, \mathcal{P}) \leq L(f, \mathcal{P}) + \varepsilon \leq \mathcal{L}(f) + \varepsilon.$$

Since this holds for all $\varepsilon > 0$, this implies $\mathcal{U}(f) - \mathcal{L}(f) \leq \varepsilon$ for all $\varepsilon > 0$ and thus must be equal, since $\mathcal{U}(f) \geq \mathcal{L}(f)$ always.

($\Rightarrow$) For any $\varepsilon > 0$, there exists a $\mathcal{P}_1$ with $L(f, \mathcal{P}_1) - \mathcal{L}(f) \leq \frac{\varepsilon}{2}$ and $\mathcal{P}_2$ with $\mathcal{U}(f) - U(f, \mathcal{P}_2) \leq \frac{\varepsilon}{2}$. Let $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2$, then the result follows by monotonicity and triangle inequality. ■

↪**Theorem 2.1**: Let $f : B \to \mathbb{R}$ be in $C^0(B)$. Then f is Riemann integrable.

Proof. Since f continuous on a compact set it is uniformly continuous. Thus for any $\varepsilon > 0$, we can find a partition $\mathcal{P}$ such that for every subbox $B_{i_1,\ldots,i_n}$, we have

$$\sup_{x \in B_{i_1,\ldots,i_n}} f(x) - \inf_{x \in B_{i_1,\ldots,i_n}} f(x) < \frac{\varepsilon}{\mathrm{vol}(B)}.$$

This readily implies $U(f, \mathcal{P}) - L(f, \mathcal{P}) \leq \varepsilon$, so the result follows by the previous lemma. ■

↪**Lemma 2.3** (Mean Value Theorem for Integrals): Let $f : B \to \mathbb{R}$ be continuous. Then, there exists $x_0 \in B$ such that $\int_B f \, \mathrm{d}V = f(x_0)\, \mathrm{vol}(B)$.

Proof. Let $m := \inf_B |f|$, $M := \sup_B |f|$. Then

$$m \cdot \mathrm{vol}(B) \leq \int_B f \, \mathrm{d}V \leq M \cdot \mathrm{vol}(B).$$

Then we can just apply the intermediate value theorem to f. ■

§2.2 General Riemann Integration

We've defined the Riemann integral for f continuous over box domains, but we wish to define the integral for more general domains. Let Ω a bounded domain in $\mathbb{R}^n$ and let B a box with $\Omega \subset B$. Extend $f : \Omega \to \mathbb{R}$ to $F : B \to \mathbb{R}$ by

$$F(x) := \begin{cases} f(x) & x \in \Omega \\ 0 & x \in B \setminus \Omega \end{cases}.$$

Heuristically, then, we wish the integral to satisfy the following:

$$\int_B f \, \mathrm{d}V = \int_\Omega f \, \mathrm{d}V + \int_{B \setminus \Omega} F \, \mathrm{d}V = \int_\Omega f \, \mathrm{d}V.$$

↪**Definition 2.7**: Given $f : \Omega \to \mathbb{R}$ bounded, set

$$\int_\Omega f \, \mathrm{d}V := \int_B F \, \mathrm{d}V,$$

whenever the right-hand side exists.

This leaves the obvious next question - what happens when f not continuous? When is the integral defined?

↪**Definition 2.8**: Let $\Gamma \subset \mathbb{R}^n$ is of *content zero* if for all $\varepsilon > 0$, there exist closed box domains $B_1^0, \ldots, B_N^0 \subset \mathbb{R}^n$ such that $\Gamma \subset \bigcup_{n=1}^N B_i^0$ and $\sum_{i=1}^N \mathrm{vol}\left(B_i^0\right) < \varepsilon$.

Remark 2.3:

1. The closed boxes can be replaced with open boxes
2. If $D_1, \ldots, D_N$ are content zero so are their union

↪**Theorem 2.2**: Suppose $f : B \subset \mathbb{R}^n \to \mathbb{R}$ bounded. Let

$$\Gamma(f) := \{x \in B : f \text{ discontinuous at } x\}.$$

Then, if $\Gamma(f)$ has content zero, then $\int_B f \, \mathrm{d}V$ exists.

Proof. We prove using the lemma following the definition of Riemann integrability. Let $\varepsilon > 0$ and put $M := \sup_B |f|$. Since $\Gamma(f)$ has content zero, there exists $B_1^0, \ldots, B_N^0 \subset \mathbb{R}^n$ such that

$$\Gamma(f) \subset \bigcup_{i=1}^N B_i^0, \quad \text{and} \quad \sum_{i=1}^N \mathrm{vol}\left(B_i^0\right) < \frac{\varepsilon}{4(M+1)}.$$

Let P_0 be a partition of B such that each subrectangle $B_{i_1,\ldots,i_n}$ is either in some B_j^0, or doesn't intersect any such B_j^0 i.e. $B_{i_1,\ldots,i_n} \cap \left(\bigcup_{i=1}^N B_i^0\right) = \varnothing$. Let A be the union of these boxes that don't intersect $\bigcup_{i=1}^N B_i^0$. A is compact, being bounded ($A \subset B$) and closed (finite union of closed sets). Thus $f|_A$ is uniformly continuous, and so there exists a refined partition $\widetilde{\mathcal{P}}$ of $\mathcal{P}$ such that for each subrectangle $Q \in \widetilde{\mathcal{P}}$ that is a subset of A,

$$\sup_Q f - \inf_Q f < \frac{\varepsilon}{2(\mathrm{vol}(B)+1)}.$$

Then

$$\begin{aligned} U\left(f, \widetilde{\mathcal{P}}\right) - L\left(f, \widetilde{\mathcal{P}}\right) &= \sum_{Q \subset A} \left(\sup_Q f - \inf_Q f\right) \mathrm{vol}(Q) + \sum_{Q \in \widetilde{\mathcal{P}}, Q \not\subset A} \left(\sup_Q f - \inf_Q f\right) \mathrm{vol}(Q) \\ &\leq \frac{\varepsilon \, \mathrm{vol}(B)}{2(\mathrm{vol}(B)+1)} + (2M) \cdot \frac{\varepsilon}{4M+1} \leq \varepsilon. \end{aligned}$$

■

⊛ **Example 2.1**: Let $\Omega \subset \mathbb{R}^{n-1}$ compact and $g : \Omega \to \mathbb{R}$ continuous. Then $\Gamma := \{(x', g(x')) : x' \in \Omega\}$ is has content zero.

↪**Corollary 2.1**: Suppose $f : B \to \mathbb{R}$ bounded and $\Gamma(f)$ the union of finitely many graphs of continuous functions over closed, bounded domains in $\mathbb{R}^{n-1}$. Then f is integrable over B.

↪**Corollary 2.2**: Let $\Omega \subset \mathbb{R}^n$ closed and bounded and suppose $\partial\Omega$ the union of finitely many graphs of C^0 functions over closed domains in $\mathbb{R}^{n-1}$. Suppose $f : \Omega \to \mathbb{R}$ bounded and $\Gamma(f) \cap \Omega$ has content zero. Then $\int_\Omega f \, \mathrm{d}V$ exists.

There is a special case of the previous result that is of practical importance. Let $\Omega \subset \mathbb{R}^n$ closed and bounded and suppose $\Omega \subset \tilde{\Omega}$ a bounded a domain. Let $G \in C^1\left(\tilde{\Omega}\right)$ be a defining function for $\partial\Omega$, i.e.

$$\partial\Omega = \left\{G(x) = 0 : x \in \tilde{\Omega}\right\}, \quad \nabla G(x) \neq 0 \, \forall \, x \in \partial\Omega.$$

By inverse-function theorem, $\partial\Omega$ is the finitely many (since compact) graphs of C^1-functions over closed and bounded domains in $\mathbb{R}^{n-1}$. Thus, we have the following:

↪**Corollary 2.3**: Let $\Omega := \{G(x) \leq 0, x \in \mathbb{R}^n\}$ be bounded and assume $\nabla G(x) \neq 0$ whenever $G = 0$. Suppose $f : \Omega \to \mathbb{R}$ bounded and $\Gamma(f) \cap \Omega$ has content zero. Then $\int_\Omega f \, \mathrm{d}V$ exists.

§2.3 Basic Properties of Integrals

↪**Proposition 2.1**: Assume $\Omega \subset \mathbb{R}^n$ bounded and $f, g : \Omega \to \mathbb{R}$ are both Riemann integrable over Ω. Then,

1. $\int_\Omega (f + g) \, \mathrm{d}V = \int_\Omega f \, \mathrm{d}V + \int_\Omega g \, \mathrm{d}V$
2. $\int_\Omega \alpha f \, \mathrm{d}V = \alpha \int_\Omega f \, \mathrm{d}V, \quad \alpha \in \mathbb{R}$
3. $f \leq g \Rightarrow \int_\Omega f \, \mathrm{d}V \leq \int_\Omega g \, \mathrm{d}V$
4. If $\Omega = \Omega_1 \cup \Omega_2$ with each Ω_i bounded with $\Omega_1 \cap \Omega_2$ having content zero, then

$$\int_\Omega f \, \mathrm{d}V = \int_{\Omega_1} f \, \mathrm{d}V + \int_{\Omega_2} f \, \mathrm{d}V.$$

§2.4 Fubini's Theorem

Given $\Omega \subset \mathbb{R}^n$ bounded etc, how do we actually compute $\int_\Omega f \, \mathrm{d}V$? Consider $B = [a, b] \times [c, d] \subset \mathbb{R}^2$ (2 dimensions is sufficient to demonstrate the main results).

Given $f : B \to \mathbb{R}$ bounded, we define the iterated integrals

$$\int_a^b \int_c^d f(x, y) \, \mathrm{d}y \, \mathrm{d}x := \int_a^b \left[\int_c^d f(x, y) \, \mathrm{d}y \right] \mathrm{d}x,$$

similarly for the other order. We assume formally for now that both 1-dimensional integrals involved exists, i.e. that $\int_a^b f(x, y) \, \mathrm{d}x$ exists for $y \in [c, d]$ and $\int_c^d f(x, y) \, \mathrm{d}y$ exists for $x \in [a, b]$. Fubini essentially says that both of these iterated integrals are equal to $\int_B f \, \mathrm{d}V$.

Remark 2.4: For $\Omega \subset \mathbb{R}^2$ we often denote $\int_\Omega f \, \mathrm{d}V =: \iint_\Omega f \, \mathrm{d}A$.

↪**Theorem 2.3** (Fubini for Box Domains): Suppose $f : B \to \mathbb{R}$ is Riemann integrable with $B = [a,b] \times [c,d]$.

1. Assume $\int_c^d f(x,y) \, \mathrm{d}y$ exists for all $x \in [a,b]$. Then the iterated integral $\int_a^b \int_c^d f(x,y) \, \mathrm{d}y \, \mathrm{d}x$ exists and equals $\int_\Omega f \, \mathrm{d}V$.
2. The same statement, but with the other iterated integral.

Proof. For any $\varepsilon > 0$ choose $\mathcal{P}$ of B such that $U(f,\mathcal{P}) - L(f,\mathcal{P}) < \varepsilon$. This partition induces partitions

$$\mathcal{P}_x = \{x_0 = a < x_1 < \cdots < x_{n-1} < x_n = b\}, \quad \mathcal{P}_y = \{y_0 = c < y_1 < \cdots < y_{m-1} < y_m = d\}.$$

Set $F(x) = \int_c^d f(x,y) \, \mathrm{d}y = \sum_{j=1}^m \int_{y_{j-1}}^{y_j} f(x,y) \, \mathrm{d}y$. We compute

$$U(F,\mathcal{P}_x) = \sum_{i=1}^n \left(\sup_{x \in [x_{i-1},x_i]} F(x) \right) \Delta x_i, \quad L(F,\mathcal{P}_x) = \sum_{i=1}^n \left(\inf_{x \in [x_{i-1},x_i]} F(x) \right) \Delta x_i.$$

Thus, we find

$$\begin{aligned}
U(F,\mathcal{P}_x) &= \sum_{i=1}^n \left(\sup_{x \in [x_{i-1},x_i]} \sum_{j=1}^m \int_{y_{j-1}}^{y_j} f(x,y) \, \mathrm{d}y \right) \Delta x_i \\
&\leq \sum_{i=1}^n \sum_{j=1}^m \left[\sup_{x \in [x_{i-1},x_i]} \int_{y_{j-1}}^{y_j} f(x,y) \, \mathrm{d}y \right] \Delta x_i \\
&\leq \sum_{i=1}^n \sum_{j=1}^m \left[\sup_{x \in [x_{i-1},x_i]} \sup_{y \in [y_{j-1},y_j]} f(x,y) \right] \Delta y_j \Delta x_i \\
&= \sum_{i=1}^n \sum_{j=1}^m \left[\sup_{(x,y) \in B_{ij}} f(x,y) \right] \Delta B_{ij}, \quad B_{ij} := [x_{i-1},x_i] \times [y_{j-1},y_j] \\
&= U(f,\mathcal{P}).
\end{aligned}$$

Similar work shows $L(F,\mathcal{P}) \geq L(f,\mathcal{P})$ and thus

$$U(F,\mathcal{P}) - L(F,\mathcal{P}) \leq U(f,\mathcal{P}) - L(f,\mathcal{P}) < \varepsilon$$

which gives the proof. ■

2.4.1 Fubini for More General Domains

↪**Definition 2.9**: A domain $\Omega \subset \mathbb{R}^2$ is called *y-simple* if it is of the form

$$\Omega = \{a \leq x \leq b, \varphi_1(x) \leq y \leq \varphi_2(x)\},$$

for some $\varphi_1, \varphi_2 \in C^0([a,b])$. Similarly it is called *x-simple* if

$$\Omega = \{c \leq y \leq d, \psi_1(y) \leq x \leq \psi_2(y)\}$$

for some $\psi_1, \psi_2 \in C^0([c,d])$. Finally, it is called *simple* if it is both x- and y-simple, and *elementary* if it is x- or y-simple.

↪**Theorem 2.4** (Fubini): Suppose $\Omega \subset \mathbb{R}^2$ is bounded and f integrable over Ω.

1. If Ω is y-simple and for all $x \in [a,b]$, $\int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)\,\mathrm{d}y$ exists, then
$$\int_\Omega f\,\mathrm{d}V = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)\,\mathrm{d}y\,\mathrm{d}x.$$
2. The analogous statement for x-simple.
3. If both x- and y-simple, we can compute the Riemann integral with either iterated integral in 1., 2..

Proof. Suppose Ω y-simple. Choose a box domain $B = [a,b] \times [c,d]$ for which $\Omega \subset B$. Then
$$\iint_\Omega f\,\mathrm{d}A = \iint_B F\,\mathrm{d}A, \quad F := \begin{cases} f & \text{in } \Omega \\ 0 & \text{else} \end{cases}.$$
Note F RI since f is. By Fubini on box domains,
$$\iint_B F\,\mathrm{d}A = \int_a^b \int_c^d F(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} F(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)\,\mathrm{d}y\,\mathrm{d}x,$$
since $F(x,\cdot)$ is zero outside $y \in [\varphi_1(x), \varphi_2(x)]$. The proof for x-simple domains follows similarly. ■

⊛ **Example 2.2**: Compute $\iint_\Omega x\,\mathrm{d}A$ where $\Omega = \{(x,y) : 0 \le x \le \sqrt{\pi}, 0 \le y \le \sin(x^2)\}$. It's clear $(x,y) \mapsto x$ continuous on Ω and since Ω is bounded and y-simple, the double integral exists and moreover can be computed using Fubini's. We have
$$\begin{aligned}\iint_\Omega x\,\mathrm{d}A &= \int_0^{\sqrt{\pi}} \int_0^{\sin(x^2)} x\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^{\sqrt{\pi}} x\sin(x^2)\,\mathrm{d}x \quad [u = x^2] \\ &= \frac{1}{2}\int_0^{\pi} \sin(u)\,\mathrm{d}u = 1.\end{aligned}$$

⊛ **Example 2.3**: Compute $I = \int_0^1 \int_y^1 e^{x^2}\,\mathrm{d}x\,\mathrm{d}y$. We can't compute the inner integral in closed-form. But, if we let $\Omega := \{(x,y) : 0 \le y \le 1, y \le x \le 1\}$, we find that $I = \iint_\Omega e^{x^2}\,\mathrm{d}A$. We can rewrite this domain as $\Omega = \{(x,y) : 0 \le x \le 1, 0 \le y \le x\}$, and thus by Fubini's (everything that needs to be continuous is continuous),
$$\begin{aligned} I &= \int_0^1 \int_0^x e^{x^2}\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^1 xe^{x^2}\,\mathrm{d}x \\ &= \frac{1}{2}(e-1).\end{aligned}$$

2.4.2 Fubini in $\mathbb{R}^n$

Given a box domain $B \subset \mathbb{R}^n$, write

$$\underline{\int_B} f \, \mathrm{d}V := \mathcal{L}(f), \quad \overline{\int_B} f \, \mathrm{d}V := \mathcal{U}(f)$$

(here, f needn't be RI).

↪**Theorem 2.5**: Let $B = B_1 \times B_2$ be a box domain in $\mathbb{R}^n$ where $B_1 \subset \mathbb{R}^k$, $B_2 \subset \mathbb{R}^{n-k}$ are also boxes. Suppose $f : B \to \mathbb{R}$ is Riemann integrable. Set

$$\ell(x) := \underline{\int_{B_2}} f(x,y) \, \mathrm{d}V(y), \quad u(x) := \overline{\int_{B_2}} f(x,y) \, \mathrm{d}V(y), \quad x \in B_1$$

and

$$\tilde{\ell}(y) := \underline{\int_{B_1}} f(x,y) \, \mathrm{d}V(x), \quad \tilde{u}(y) := \overline{\int_{B_1}} f(x,y) \, \mathrm{d}V(x), \quad y \in B_2.$$

Then, $\ell(x)$ and $u(x)$ are Riemann integrable over B_1 and $\tilde{\ell}(y)$ and $\tilde{u}(y)$ are Riemann integrable over B_2. Moreover,

$$\int_B f \, \mathrm{d}V = \int_{B_1} \ell(x) \, \mathrm{d}V(x) = \int_{B_1} u(x) \, \mathrm{d}V(x),$$
$$\int_B f \, \mathrm{d}V = \int_{B_2} \tilde{\ell}(y) \, \mathrm{d}V(y) = \int_{B_2} \tilde{u}(y) \, \mathrm{d}V(y).$$

Proof. The proof follows in the same way as in $\mathbb{R}^2$ by noting the following:

$$L(f, \mathcal{P}) \leq L(\ell(x), \mathcal{P}_x), \quad U(u(x), \mathcal{P}_x) \leq U(f, \mathcal{P}).$$

This implies, suping/infing over all $\mathcal{P}$,

$$\mathcal{L}(f) \leq \mathcal{L}(\ell(x)), \quad \mathcal{U}(u(x)) \leq \mathcal{U}(f)$$

and thus since f Riemann integrable,

$$0 = \mathcal{U}(f) - \mathcal{L}(f) \geq \mathcal{U}(u(x)) - \mathcal{L}(\ell(x)) \geq 0 \Rightarrow \mathcal{U}(u(x)) = \mathcal{L}(\ell(x))$$

thus $\ell(x)$ Riemann integrable and in particular $\mathcal{L}(f) = \mathcal{L}(\ell(x)) = \mathcal{U}(\ell(x)) = \mathcal{U}(f)$. Similar holds in the y-variables. ■

Remark 2.5: If for all $x \in B_1, f(x,y)$ is integrable in B_2 (i.e. $\int_{B_2} f(x,y) \, \mathrm{d}V(y)$ exists), then

$$\int_B f \, \mathrm{d}V = \int_{B_1} \left[\int_{B_2} f(x,y) \, \mathrm{d}V(y) \right] \mathrm{d}V(x).$$

Similarly, if $\int_{B_1} f(x,y) \, \mathrm{d}V(x)$ exists for all $y \in B_2$, then

$$\int_B f \, \mathrm{d}V = \int_{B_2} \left[\int_{B_1} f(x,y) \, \mathrm{d}V(x) \right] \mathrm{d}V(y).$$

2.4.3 Fubini's in $\mathbb{R}^3$

↪**Definition 2.10**: A bounded domain $W \subset \mathbb{R}^3$ is called *z-simple* if there exists an elementary region $\Omega \subset \mathbb{R}^2$ and $\eta_1, \eta_2 \in C^0(\Omega)$ such that

$$W = \{(x,y,z) : (x,y) \in \Omega, \eta_1(x,y) \leq z \leq \eta_2(x,y)\}.$$

Analogous definitions hold for x-, y-simple as in the $\mathbb{R}^2$ case. We additionally say W elementary if it is simple in some axis.

↪**Theorem 2.6** (Fubini in $\mathbb{R}^3$): Suppose $W \subset \mathbb{R}^3$ is bounded and z-simple and $f : W \to \mathbb{R}$ is RI. Assume that for all $(x,y) \in \Omega$, $\int_{\eta_1(x,y)}^{\eta_2(x,y)} f(x,y,z)\,\mathrm{d}z$ exists. Then, it is Riemann integrable over Ω and

$$\iiint_W f\,\mathrm{d}V = \int_\Omega \left[\int_{\eta_1(x,y)}^{\eta_2(x,y)} f(x,y,z)\,\mathrm{d}z\right] \mathrm{d}V(x,y).$$

Similar holds for x-, y- simple domains.

Proof. Follows as in $\mathbb{R}^2$. ■

§2.5 Change of Variables Formula

Let $\Omega \subset \mathbb{R}^n$ be a bounded domain. We defined $\int_\Omega f\,\mathrm{d}V$ by looking at $L(f,\mathcal{P})$ and $U(f,\mathcal{P})$ where $\mathcal{P}$ was a *rectangular partition*. Can we use other partitions of Ω that are *not* rectangular?

↪**Definition 2.11**: Let $\Omega \subset \mathbb{R}^n$ be a domain with content zero boundary. A *regular partition* $\mathcal{P}_{\text{reg}} = \{D_k\}_{k=1}^N$ is a collection of subdomains of $D_k \subset \Omega$ such that
1. $\Omega = \bigcup_{k=1}^N D_k$ and $\text{Content}(\partial D_k) = 0$ for all $k = 1, \ldots, N$, and
2. $\text{Content}(D_k \cap D_\ell) = 0$ for $k \neq \ell$.

↪**Definition 2.12**: $f : \Omega \to \mathbb{R}$ is bounded where Ω a bounded domain and $\mathcal{P}_{\text{reg}} = \{D_k\}_{k=1}^N$ a regular partition of Ω. Define

$$U(f,\mathcal{P}_{\text{reg}}) := \sum_{k=1}^N \left(\sup_{D_k} f\right) \text{vol}(D_k), \quad L(f,\mathcal{P}_{\text{reg}}) := \sum_{k=1}^N \left(\inf_{D_k} f\right) \text{vol}(D_k).$$

↪**Lemma 2.4**: Let $f : B \to \mathbb{R}$. Then f is Riemann integrable over B iff for any $\varepsilon > 0$ there exists a regular partition $\mathcal{P}_{\text{reg}}$ of B such that

$$U(f,\mathcal{P}_{\text{reg}}) - L(f,\mathcal{P}_{\text{reg}}) < \varepsilon.$$

Proof. ($\Rightarrow$) is immediate, since a rectangular partition is regular and we already know the analogous statement for such partitions.

($\Leftarrow$) Fix $\varepsilon > 0$ and let $\mathcal{P}_{\text{reg}} = \{D_k\}$ is such that

$$U(f,\mathcal{P}_{\text{reg}}) - L(f,\mathcal{P}_{\text{reg}}) < \varepsilon.$$

Then, $\Gamma := \bigcup_k \partial D_k$ has content zero by assumption, and so there exist boxes $B_1, \ldots, B_m$ such that $\Gamma \subset \bigcup_j B_j$ and $\sum_j \operatorname{vol}\left(B_j\right) < \frac{\varepsilon}{2(M+1)}$ where $M := \sup_B |f|$. Pick a box partition $\mathcal{P}$ such that

$$\sum_{\substack{B_i \in \mathcal{P} \\ \mathcal{B}_i \cap \Gamma \neq \varnothing}} \operatorname{vol}(B_i) < \frac{\varepsilon}{2(M+1)},$$

and for each $B_j \subset \mathcal{P}$ with $B_j \cap \Gamma = \varnothing$ we have that $B_j \subset D_\ell$ for some ℓ (obtained by refining). Note that if B_j is such that $B_j \cap \Gamma$,

$$0 \leq \sup_{B_j} f - \inf_{B_j} f \leq \sup_{D_\ell} f - \inf_{D_\ell} f.$$

Then

$$\begin{aligned}
U(f, \mathcal{P}) - L(f, \mathcal{P}) &= \sum_{\substack{B_i \in \mathcal{P} \\ B_i \cap \Gamma \neq \varnothing}} \left(\sup_{B_i} f - \inf_{B_i} f \right) \operatorname{vol}(B_i) + \sum_{\substack{B_i \in \mathcal{P} \\ B_i \cap \Gamma = \varnothing}} \left(\sup_{B_i} f - \inf_{B_i} f \right) \operatorname{vol}(B_i) \\
&\leq 2M \sum_{\substack{B_i \in \mathcal{P} \\ B_i \cap \Gamma \neq \varnothing}} \operatorname{vol}(B_i) + \sum_{D_\ell} \left(\sup_{D_\ell} f - \inf_{D_\ell} \right) \operatorname{vol}(D_\ell) \\
&< (2M) \frac{\varepsilon}{2(M+1)} + U\left(f, \mathcal{P}_{\text{reg}}\right) - L\left(f, \mathcal{P}_{\text{reg}}\right) \\
&< 2\varepsilon.
\end{aligned}$$

■

↪**Theorem 2.7** (Change of Variables): Let $\Omega, \Omega' \subset \mathbb{R}^n$ bounded domains and let $T : \Omega' \to \Omega$ be a C^1 map that is bijective from Ω' onto Ω. Then, for any $f : \Omega \to \mathbb{R}$ which is Riemann-integrable over Ω, then

$$\int_\Omega f \, \mathrm{d}V = \int_{\Omega'} (f \circ T) |\det DT| \, \mathrm{d}V.$$

To prove, we need to first observe how the volume of a box B changes under application of T. Let $B' \subset \Omega'$ be a small rectangular box. Since T C^1, by Taylor expansion, we have for any $y_0 \in B' \subset \mathbb{R}^n$,

$$T(y) = T(y_0) + DT(y_0)(y - y_0) + E_1(y, y_0)$$

where $\lim_{y \to y_0} \frac{\|E_1(y, y_0)\|}{\|y - y_0\|} = 0$. Since $DT(y_0)$ is linear,

$$\operatorname{vol}(T(B')) = |\det(DT(y_0))| \operatorname{vol}(B') + E_2(B')$$

where for any $\varepsilon > 0$, there exists $\delta > 0$ such that $\|E_1(y, y_0)\| \leq \varepsilon \|y - y_0\|$ whenever $0 < \|y - y_0\| < \delta$. Suppose $B' = \left\{ y \in \mathbb{R} : \max_{j=1}^n \left| y_j - y_j^0 \right| \leq \delta \right\}$ (a box of side length δ). Using the fact that $\|y - y_0\| \leq \sqrt{n} \max_{j=1}^n \left| y_j - y_j^0 \right|$. Then,

$$E_2(B') \leq (\varepsilon \|y - y_0\|)^n \leq \left(\varepsilon \sqrt{n} \max_j \left| y_j - y_j^0 \right| \right)^n \leq (\sqrt{n})^n \varepsilon^n \left(\max_j \left| y_j - y_j^0 \right| \right)^n = \left(\sqrt{n}^n \varepsilon^n \right) \operatorname{vol}(B') \leq C_n \varepsilon \operatorname{vol}(B'),$$

where C_n some dimensional constant. Thus,

$$\frac{|E_2(B')|}{\operatorname{vol}(B')} \to 0 \text{ as } \|y - y_0\| \to 0.$$

Remark that since $T \in C^1(B')$, $|\det(DT)| \in C^0(B')$ and so uniformly continuous in B', so given $\varepsilon > 0$, we can find $\delta > 0$ such that $|E_2(B')| \leq \varepsilon \operatorname{vol}(B')$ and $\|\det(DT(y_0))| - |\det DT(y)\| \leq \varepsilon$ for any $y \in B'$ with $\|y - y_0\| < \delta$ (?) and thus, for all $\varepsilon > 0$, there exists a B' of sidelength $\leq \delta$,

$$\operatorname{vol}(TB') = |\det DT(y)| \operatorname{vol}(B') + \tilde{E}(B'), \quad |\tilde{E}(B')| \leq \varepsilon \operatorname{vol}(B')$$

Proof. We use the observations above and the previous lemma on regular partitions. For any $\varepsilon > 0$, we can find $\delta > 0$ and a box B' with sidelength $< \delta$ such that

$$\operatorname{vol}(TB') = |\det DT(y)| \operatorname{vol}(B') + E_2(B'),$$

for any $y \in B'$, where $|E_2(B')| \leq \varepsilon \operatorname{vol}(B')$.

Choose a box partition $\mathcal{P}' = \{B'_j\}$ on Ω'. Since $T : \Omega' \to \Omega$ a C^1 bijection onto Ω, $\mathcal{P}_{\text{reg}} = \{D_i = T(B'_i)\}$ is a regular partition on Ω (why?). We assume that $U(f, \mathcal{P}_{\text{reg}}) - L(f, \mathcal{P}_{\text{reg}}) < \varepsilon$ and $|E_2(B'_i)| \leq \varepsilon \operatorname{vol}(B'_i)$ for all $B'_i \in \mathcal{P}'$, by refinement and since f integrable. Then,

$$U(f, \mathcal{P}_{\text{reg}}) \geq U((f \circ T)|\det(DT)|, \mathcal{P}') - 2M\varepsilon \operatorname{vol}(\Omega'),$$

and similarly

$$L(f, \mathcal{P}_{\text{reg}}) \leq L((f \circ T)|\det(DT)|, \mathcal{P}') + 2M\varepsilon \operatorname{vol}(\Omega')$$

where $M = \sup_\Omega |f|$. This implies

$$\begin{aligned}\varepsilon > U(f, \mathcal{P}_{\text{reg}}) - L(f, \mathcal{P}_{\text{reg}}) &\geq U((f \circ T)|\det(DT)|, \mathcal{P}') - L((f \circ T)|\det(DT)|, \mathcal{P}') - 4M\varepsilon \operatorname{vol}(\Omega') \\ &\Rightarrow U((f \circ T)|\det(DT)|, \mathcal{P}') - L((f \circ T)|\det(DT)|, \mathcal{P}') < \varepsilon(1 + 4M \operatorname{vol}(\Omega')).\end{aligned}$$

Since ε arbitrary, this proves the result. ■

⊛ **Example 2.4** (Polar Coordinates): Take $x = r\cos\theta, y = r\sin\theta$, so $r = \sqrt{x^2 + y^2}$. Let $T(r, \theta) = (x, y)$ from $\Omega' \to \Omega$ with $\Omega' = \{(r, \theta) \in (0, a] \times [0, 2\pi)\}$ and $\Omega = \{(x, y) : x^2 + y^2 \leq a^2\}$. Then

$$\iint_{\{x^2+y^2 \leq a^2\}} f(x, y)\, \mathrm{d}A = \iint_{\Omega'} f(r\cos(\theta), r\sin(\theta))|\det(DT)|\, \mathrm{d}r\, \mathrm{d}\theta,$$

where

$$\det(DT) = \det\left(\begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}\right) = r.$$

By Fubini, then, we get

$$\iint_{\{x^2+y^2 \leq a^2\}} f(x, y)\, \mathrm{d}A = \int_0^{2\pi} \int_0^a f(r\cos(\theta), r\sin(\theta)) r\, \mathrm{d}r\, \mathrm{d}\theta.$$

⊛ **Example 2.5** (Spherical Coordinates): Let $x = \rho \sin\varphi \cos\theta, y = \rho \sin\varphi \sin\theta, z = \rho\cos\varphi$ and $T(\rho,\varphi,\theta) = (x,y,z)$ with $\theta \in [0,2\pi), \varphi \in [0,\pi)$ and $(0,a]$. In this case

$$|\det(DT)| = |\det\begin{pmatrix} \sin\varphi\cos\theta & \rho\cos\varphi\cos\theta & -\rho\sin\varphi\sin\theta \\ \sin\varphi\sin\theta & \rho\cos\varphi\sin\theta & \rho\sin\varphi\cos\theta \\ \cos\varphi & -\sin\varphi & 0 \end{pmatrix}| = \rho^2\sin\varphi$$

so that

$$\iiint_{\{x^2+y^2+z^2\le a^2\}} f(x,y,z)\,\mathrm{d}V = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} f(\rho\sin\varphi\cos\theta, \rho\sin\varphi\sin\theta, \rho\cos\varphi).$$

⊛ **Example 2.6**: Compute $\iint_\Omega (y-x)\cos(xy)\,\mathrm{d}A$, with Ω the region bounded by $x+y=1, x+y=4, xy=-1$ and $xy=-4$ and $y-x\ge 0$. Let $u = x+y$, $v = xy$ and $T(u,v) = (x,y)$ with $T : \Omega' \to \Omega$, where we now have $\Omega' = \{(u,v) \in [1,4]\times[-4,-1]\}$. Moreover, we have

$$(x+y)^2 - (x-y)^2 = 4xy \Rightarrow u^2 - (x-y)^2 = 4v \Rightarrow |x-y| = \sqrt{u^2-4v}.$$

Finally,

$$|\det(DT^{-1})| = |\det\left(\begin{pmatrix} 1 & 1 \\ y & x \end{pmatrix}\right)| = |x-y| \Rightarrow |\det DT| = \frac{1}{|x-y|}.$$

This implies

$$\iint_\Omega (x-y)\cos(xy)\,\mathrm{d}A = \int_1^4\int_{-4}^{-1}\cos(v)\,\mathrm{d}v\,\mathrm{d}u = 3[\sin(-1)-\sin(-4)].$$

§3 Vector Fields, Curves and Surfaces

§3.1 Integration along Curves

↪**Definition 3.1**: A curve in $\mathbb{R}^n$ is a continuous function $\gamma : I \to \mathbb{R}^n$ where $I \subset \mathbb{R}$ some interval.

↪**Definition 3.2**: Let γ be a C^1 curve defined on $I = [0,L]$. Define the integral over γ of a real-valued function as

$$\int_\gamma f\,\mathrm{d}s := \int_0^L f(\gamma(t))|\gamma'(t)|\,\mathrm{d}t.$$

§3.2 Integration of Vector Fields

↪**Definition 3.3**: A *vector field* on $\mathbb{R}^n$ is a function $F : \Omega \subset \mathbb{R}^n \to \mathbb{R}^n$. We usually assume F C^0 on Ω.

How can we integrate a vector field along a path γ? (think: the work done by F along γ)

Let $\gamma : [a,b] \to \mathbb{R}^n$ be a C^1 path, and consider

$$\Delta s_i := \gamma(t_i + \Delta t_i) - \gamma(t_i)$$

for Δt_i *small*. This is a vector.

Thus, since γ is C^1,

$$\Delta s_i = \gamma'(t_i)\Delta t_i + R(t, \Delta t_i),$$

where $\frac{\|R(t,\Delta t_i)\|}{\Delta t_i} \to 0$ as Δt_i. We get:

$$F(\gamma(t)) \cdot \Delta s_i = F(\gamma(t_i))[\gamma'(t_i)\Delta t_i + R(t_i, \Delta t_i)].$$

Takeing $\mathcal{P} = \{t_i\}_{i=1}^N$ of $[a, b]$, we can repeat this over every partition. This motivates:

↪**Definition 3.4**: Define the *total work done* by F along γ by

$$\int_\gamma (F) \cdot (\mathrm{d}s) := \lim_{N\to\infty} \sum_{i=1}^N (F)(\gamma(t_i))(\Delta s_i),$$

as defined above. This is equal to

$$\int_a^b (F)(\gamma(t)) \cdot \gamma'(t)\,\mathrm{d}t,$$

assuming the Riemann integral exists, where γ lives on $[a, b]$.

↪**Definition 3.5**: Let γ a piecewise smooth path and F a vector field defined in a neighborhood of γ. The *line integral* (also called *path integral*) of F along γ is

$$\int_\gamma F \cdot \mathrm{d}s = \int_c^b F(\gamma(t)) \cdot \gamma'(t)\,\mathrm{d}t.$$

If γ is regular, i.e. $\gamma'(t) \neq 0$, let $T(t) = \frac{\gamma'(t)}{\|\gamma'(t)\|}$ be the *unit tangent field*. Then, we see that

$$\int_\gamma F \cdot \mathrm{d}s = \int_\gamma F(\gamma(t)) \cdot T(t),$$

where $\mathrm{d}s \equiv \|\gamma'(t)\|\,\mathrm{d}t$ the "arclength element".

Formally, we write $\operatorname{vec} \mathrm{d}s = \operatorname{vec} T\,\mathrm{d}s$.

↪**Lemma 3.1**: If γ a piecewise-smooth path and $\tilde{\gamma}$ a reparametrization of γ, then if F a C^0 vector field, then

$$\int_\gamma F \cdot \mathrm{d}s = \varepsilon \int_{\tilde{\gamma}} F \cdot \mathrm{d}s,$$

where $\varepsilon = +1$ if an orientation-preserving reparametrization, and $\varepsilon = -1$ if an orientation-reversing reparametrization.

3.2.1 Conservative Vector Fields

↪**Definition 3.6**: A vector field $F : \Omega \to \mathbb{R}^n$ is *conservative* in Ω if there exists a C^1 function $f : \Omega \to \mathbb{R}$ such that $F(x) = \nabla f(x)$ for all $x \in \Omega$.

↪**Theorem 3.1**: Suppose F a conservative vector field in a neighborhood of a piecewise-smooth path γ. Then

$$\int_\gamma F \cdot \mathrm{d}s = f(\gamma(b)) - f(\gamma(a)).$$

↪**Corollary 3.1**: If γ closed, then $\int_\gamma F \cdot \mathrm{d}s = 0$.

Proof. (Of Theorem)

$$\begin{aligned}\int_\gamma F \cdot \mathrm{d}s &= \int_a^b \nabla f(\gamma(t)) \cdot \gamma'(t)\,\mathrm{d}t \\ &= \int_a^b \frac{\mathrm{d}}{\mathrm{d}t} f(\gamma(t))\,\mathrm{d}t \\ &= f(\gamma(b)) - f(\gamma(a)).\end{aligned}$$

■

If $C = \cup_i C_i$ a finite union of curves C_i, we extend our definition to

$$\int_C F \cdot \mathrm{d}s = \sum_i \int_{C_i} F \cdot \mathrm{d}s.$$

3.2.2 Poincaré Lemma

The question we aim to address here is, when is $F : \Omega \to \mathbb{R}^n$ conservative?

A necessary condition is as follows.

↪**Theorem 3.2**: Let $F = (F_1, \ldots, F_n)$ a C^1 conservative vector field. Then,

$$\frac{\partial F_i}{\partial x_j}(x) = \frac{\partial F_j}{\partial x_i}(x)$$

for all i, j and $x \in \Omega$.

Proof. We have $F = \nabla f$ for $f \in C^1(\Omega)$; then $\frac{\partial F_i}{\partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$, so the claim follows by Clairault's. ■

When is this condition sufficient?

§3.3 Poincaré Lemma

↪**Lemma 3.2**: Suppose $g(x, y)$ is C^1 in $\Omega \times A$ where $\Omega \subset \mathbb{R}^n$ open and $A \subset \mathbb{R}^m$ is compact. Then,

$$\frac{\partial}{\partial x_i}\left(\int_A g(x,y)\,\mathrm{d}y\right) = \int_A \frac{\partial}{\partial x_i} g(x,y)\,\mathrm{d}y$$

for any $i \in [n]$ and $x \in \Omega$.

Proof. Fix $r > 0$ so that $B_r(x) \subset \Omega$. Let $h \in \mathbb{R}$ with $|h| < r$. By the mean-value theorem,

$$g(x+he_i,y)-g(x,y)=\frac{\partial g}{\partial x_i}\left(x^h,y\right)\cdot h,\quad x^h\in B_r(x).$$

Since $\overline{B_r(x)}\times A$ is compact, $\frac{\partial g}{\partial x_i}$ is uniformly continuous on this set. Thus for any $\varepsilon>0$, there exists $\delta>0$ such that

$$\left|\partial_{x_i}g\left(x^h,y\right)-\partial_{x_i}g(x,y)\right|<\varepsilon,\quad \forall\,|h|<\delta, y\in A.$$

Then,

$$\begin{aligned}\left|\overbrace{\frac{\int_A g(x+he_i,y)\,\mathrm{d}y-\int_A g(x,y)\,\mathrm{d}y}{h}}^{\text{difference quotient for integral}}-\int_A \partial_{x_i}g(x,y)\,\mathrm{d}y\right| &\leq \int_A\left|\frac{\partial g}{\partial x_i}\left(x^h,y\right)-\frac{\partial g}{\partial x_i}(x,y)\right|\mathrm{d}y\\ &\leq \varepsilon\,\mathrm{vol}(A),\end{aligned}$$

for all $\varepsilon>0$ with $|h|<\delta$. Since ε arbitrary, this proves the claim. ■

This tells us when we can differentiate under an integral sign.

↪**Definition 3.7** (Star-Shaped Region): $\Omega\subset\mathbb{R}^n$ *star-shaped* if there exists $p_0\in\Omega$, which we call a *center*, such that the line segment $\ell(p_0,x)$ connecting p_0 to x is contained entirely within Ω for all $x\in\Omega$.

↪**Theorem 3.3** (Poincaré Lemma): Suppose $\Omega\subset\mathbb{R}^n$ is star-shaped and F is a $C^1(\Omega,\mathbb{R}^n)$ vector field. Then, F is conservative iff

$$\frac{\partial F_i}{\partial x_j}=\frac{\partial F_j}{\partial x_i}$$

for all $i,j=1,\ldots,n$.

Remark 3.1: The statement holds for more general, so-called "contractible domains", but we limit our exposition here.

Proof. We know the "$\Rightarrow$" direction already. Conversely, suppose wlog that $p_0=0$ is a "center" of our star-shaped domain. The idea to construct our f such that $F=\nabla f$ is to integrate F along the straight-line curve connecting x to 0 for each x. Formally, define $\gamma_x(t)=tx, 0\leq t\leq 1$, and define

$$f(x)=\int_{\gamma_x}F\cdot\mathrm{d}s=\int_0^1 F(tx)\cdot x\,\mathrm{d}t=\sum_{i=1}^n\int_0^1 F_i(tx)x_i\,\mathrm{d}t.$$

We need to compute the gradient of this function. Since $(t,x)\mapsto F_i(tx)x_i$ clearly C^1 and $[0,1]$ compact, we can apply the lemma and differentiate in x under the integral sign, and we find

$$
\begin{aligned}
\frac{\partial f}{\partial x_j}(x) &= \int_0^1 \sum_{i=1}^n \left[\frac{\partial F_i}{\partial x_j}(tx) t x_i + F_i(tx)\delta_{i,j} \right] \mathrm{d}t \\
&= \int_0^1 F_j(tx) + \sum_{i=1}^n t x_i \frac{\partial F_i}{\partial x_j}(tx)\, \mathrm{d}t \\
\text{(integrate by parts on first summand)} \quad &= \int_0^1 \sum_{i=1}^n t x_i \frac{\partial F_i}{\partial x_j}(tx)\, \mathrm{d}t + \left[t F_j(tx) \right]_0^1 - \int_0^1 t \sum_{i=1}^n \frac{\partial F_j}{\partial x_i} x_i(tx)\, \mathrm{d}t \\
&= \int_0^1 \sum_{i=1}^n t x_i \underbrace{\left[\frac{\partial F_i}{\partial x_j}(tx) - \frac{\partial F_j}{\partial x_i}(tx) \right]}_{=0 \text{ by assumption}} \mathrm{d}t + F_j(x) = F_j(x),
\end{aligned}
$$

as claimed. ■

§3.4 Integration on Surfaces

↪**Definition 3.8**: A parametrized surface $S \subset \mathbb{R}^3$ is the image of a map $\Phi : D \to S$,

$$S = \{(x(u,v), y(u,v), z(u,v)) = \Phi(u,v) : (u,v) \in D\},$$

where $D \subset \mathbb{R}^2$ some bounded domain. If Φ is C^k we call S a C^k surface. Note that

$$\Phi_u = (\partial_u x, \partial_u y, \partial_u z), \quad \Phi_v = (\partial_v x, \partial_v y, \partial_v z)$$

are tangent to S.

↪**Definition 3.9**: For a regular C^1 parametrization $\Phi : D \to S$, define

$$\operatorname{area}(S) := \iint_D \|\Phi_u \times \Phi_v\| \, \mathrm{d}A,$$

when it exists.

↪**Definition 3.10**: Let $S = \Phi(D)$ a C^1 parametrized surface and suppose $f \in C^0(S)$. Then we define

$$\iint_S f \, \mathrm{d}S := \iint_D f(\Phi) \|\Phi_u \times \Phi_v\| \, \mathrm{d}A.$$

↪**Proposition 3.1**: In the special case that S is the graph of a C^1 function $g : D \to \mathbb{R}$ (i.e. $S = \{(x,y,g(x,y)) : (x,y) \in D\}$), then this integral becomes

$$\iint_D f(x,y,g(x,y)) \sqrt{1 + g_x^2 + g_y^2} \, \mathrm{d}A.$$

Proof. This follows from computing

$$\Phi_x = (1,0,g_x), \quad \Phi_y = \left(0,1,g_y\right) \Rightarrow \Phi_x \times \Phi_y = \left(-g_x, -g_y, 1\right).$$

■

↪**Definition 3.11**: Given $S = \Phi(D)$ a C^1 parametrized surface, we say S is regular if the normal $\Phi_u \times \Phi_v$ never vanishes over D, i.e. if Φ_u, Φ_v are never linearly dependent. If S regular, we can choose

$$N = \pm \frac{\Phi_u \times \Phi_v}{\|\Phi_u \times \Phi_v\|}$$

to be the *unit normal*; there are two choices of sign, leading to different *orientations* of S.

↪**Definition 3.12**: Let F a C^0 vector field on a regular parametrized surface $S = \Phi(D)$. Then, we define

$$\iint_S F \cdot \mathrm{d}s := \iint_D F \cdot N \|\Phi_u \times \Phi_v\| \,\mathrm{d}u \,\mathrm{d}v = \iint_D (F \circ \Phi) \cdot (\Phi_u \times \Phi_v) \,\mathrm{d}A.$$

Note that this is defined up to orientation of S; however, it isn't clear from the definition that this is independent of parametrization.

↪**Definition 3.13**: Given $\Phi : D \to S, \widetilde{\Phi} : \tilde{D} \to S$ regular C^1 parametrizations of the same surface, we say $\widetilde{\Phi}$ is a reparametrization of Φ if there exists a bijective C^1 map $h : \tilde{D} \to D$ with C^1 inverse, such that $\widetilde{\Phi}(s,t) = (\Phi \circ h)(s,t)$.

We say $\widetilde{\Phi}$ *orientation-preserving* if

$$\frac{(\Phi_u \times \Phi_v)(h(s,t))}{\|(\Phi_u \times \Phi_v)(h(s,t))\|} = \frac{\left(\widetilde{\Phi}_s \times \widetilde{\Phi}_t\right)(s,t)}{\left\|\left(\widetilde{\Phi}_s \times \widetilde{\Phi}_t\right)(s,t)\right\|},$$

and *reversing* if the same equality holds with a negative in front of the second term.

↪**Lemma 3.3**: Let $\widetilde{\Phi}, \Phi$ as above with $h : \tilde{D} \to D$ written as $h(s,t) = (u(s,t), v(s,t)) \in C^1(\tilde{D}, D)$ a change of parameters. Then h is orientation-preserving iff $\det\left(\frac{\partial(u,v)}{\partial(s,t)}\right) > 0$, and reversing iff $\det\left(\frac{\partial(u,v)}{\partial(s,t)}\right) < 0$, where $\frac{\partial(u,v)}{\partial(s,t)}$ the Jacobian matrix of h.

Proof. Follows from the chain rule upon computing $\widetilde{\Phi}_s$ and $\widetilde{\Phi}_t$, namely we find

$$\widetilde{\Phi}_s \times \widetilde{\Phi}_t = \det\left(\frac{\partial(u,v)}{\partial(s,t)}\right)(\Phi_u \times \Phi_v),$$

from which the claim readily follows. ■

From this lemma, it readily follows that the previous definition of the integral of a vector field over a surface is well-defined (modulo orientation) regardless of choice of parametrization, by applying the chain of variables formula and the previous lemma. We'll often denote a positively/negatively oriented surface by S^+, S^- depending on the choice of sign of the normal vector. As an immediate corollary, we find

$$\iint_{S^\pm} F \cdot \mathrm{d}s = -\iint_{S^\mp} F \cdot \mathrm{d}s.$$

§3.5 Green's Theorem and its Consequences

↪**Definition 3.14**: A surface $S \subset \mathbb{R}^3$ is said to be a *regular surface with C^1 boundary* ∂S if S is compact and if for all $p \in S$, there exists $r, \delta > 0$ with $B_r(\rho) \subset \mathbb{R}^3$ and $D = \{-\delta < u < \delta, -\delta < v < \delta\} \subset \mathbb{R}^2$ and a regular parametrization map $\Phi : D \to \mathbb{R}^3$ such that either

1. $S \cap B_r(\rho) = \Phi(D)$, or
2. $S \subset B_r(\rho) = \Phi(D^+)$ where $D_+ := \{-\delta < u < \delta, 0 \leq v < \delta\}$. Points in 1. are called *interior charts* of S and points in 2. are called *boundary charts of S*; the boundary of S is defined as $\partial S = \{\text{all boundary points of } S\}$. The local parametrized surfaces are called coordinate charts.

↪**Definition 3.15**: A regular surface $S \subset \mathbb{R}^3$ is orientable if the unit normal $n \in C^0(S, \mathbb{R}^3)$.

In general there are two choices of unit normal at any point of a surface of $\mathbb{R}^3$ ("in/out"). Given a regular orientable surface $S \subset \mathbb{R}^3$, we say it is positively oriented if:

1. If $\partial S = \varnothing$, the unit normal is pointing "outwards" towards the "unbounded" region of the ambient $\mathbb{R}^3$
2. If $\partial S \neq \varnothing$, then we choose a "positive/counterclockwise" orientation for the curve ∂S which induces an orientation on S by the right-hand rule.

We are now in the position to state and prove Green's, which relates line integrals of vector fields over ∂D to double integrals over a planar domain D. We say that a bounded planar domain $D \subset \mathbb{R}^2$ is positive if the unit normal (with respect to orientation on boundary) is $\lambda = (0, 0, 1)$ (equivalently, we can embed D into $\mathbb{R}^3$ as a (planar) surface, then positive orientation is the same as that defined above for surfaces).

↪**Theorem 3.4** (Green): Let $D \subset \mathbb{R}^2$ be a positively oriented bounded piecewise C^1 domain and $F = (P, Q) : D \to \mathbb{R}^2$ a vector field on D. Then

$$\int_{\partial D} F \cdot \mathrm{d}s = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathrm{d}A.$$

Proof. On ∂D, we have the short-hand $\mathrm{d}s = i\,\mathrm{d}x + j\,\mathrm{d}y = (\mathrm{d}x, \mathrm{d}y)$ where $i = (1, 0), j = (0, 1)$. We can write

$$\begin{aligned} \int_{\partial D} F \cdot \mathrm{d}s &= \int_{\partial D} (P, Q) \cdot (\mathrm{d}x, \mathrm{d}y) \\ &= \int_{\partial D} P\,\mathrm{d}x + Q\,\mathrm{d}y. \end{aligned}$$

Assume first D is a simple domain

$$D = \{a \leq x \leq b, \varphi_1(x) \leq y \leq \varphi_2(x)\} = \{c \leq y \leq d, \psi_1(y) \leq x \leq \psi_2(y)\}$$

with boundary C. We claim $-\iint_D \frac{\partial P}{\partial y}\,\mathrm{d}A = \int_C P\,\mathrm{d}x$. Write

$$
\begin{aligned}
C_1 &= \{a \le x \le b, y = \varphi_1(x)\} \\
C_2 &= \{a \le x \le b, y = \varphi_2(x)\} \\
B_1 &= \{x = a, \varphi_1(a) \le y \le \varphi_2(a)\} \\
B_2 &= \{x = b, \varphi_1(b) \le y \le \varphi_2(b)\}.
\end{aligned}
$$

C_1^+, C_2^+ are oriented $a \mapsto b$ and B_1^+, B_2^+ are oriented from $\varphi_1(a)$ to $\varphi_2(a), \varphi_1(b)$ to $\varphi_2(b)$ respectively. By Fubini,

$$
\begin{aligned}
\iint_D \frac{\partial P}{\partial y}\,\mathrm{d}A &= \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} \frac{\partial P}{\partial y}\,\mathrm{d}y\,\mathrm{d}x \\
\text{(FTC)} \quad &= \int_a^b (P(x,\varphi_2(x)) - P(x,\varphi_1(x)))\,\mathrm{d}x \\
&= \int_{C_2^+} P\,\mathrm{d}x - \int_{C_1^+} P\,\mathrm{d}x = \int_{C_2^+ \cup C_{\bar{1}}} P\,\mathrm{d}x \\
&= -\int_{C_{\bar{2}} \cup C_1^+} P\,\mathrm{d}x + \int_{B_1^+} P\,\mathrm{d}x - \int_{B_2^+} P\,\mathrm{d}x \\
&= -\int_{C_{\bar{2}} \cup C_1^+ \cup (B_2^+) \cup B_{\bar{1}}} P\,\mathrm{d}x \\
&= -\int_C P\,\mathrm{d}x,
\end{aligned}
$$

where we use the fact that

$$
\int_{B_1^+} P\,\mathrm{d}x - \int_{B_2^+} P\,\mathrm{d}x = 0
$$

since x is constant on B_1 and B_2 to add them in to our computations.

Similarly, one shows that

$$
\iint_D \frac{\partial Q}{\partial x}\,\mathrm{d}A = \int_{\partial D} Q\,\mathrm{d}y.
$$

Combining these identities gives the proof.

For a more general domain, one case use the implicit function theorem and the piecewise-C^1 assumption to reduce the domain to a union of simple regions, which sum up to give the general result. ■

↪**Corollary 3.2**:

$$
\int_{\partial D} x\,\mathrm{d}y = -\int_{\partial D} y\,\mathrm{d}x = \frac{1}{2}\int_{\partial D} x\,\mathrm{d}y - y\,\mathrm{d}x = \operatorname{area}(D),
$$

where we use as in the previous theorem the notation, where if $\gamma = (\gamma_1, \gamma_2) : I \to \mathbb{R}^2$ is a parametrization of ∂D, then

$$
\mathrm{d}x = \gamma_1'(t)\,\mathrm{d}t,
$$

etc (and the input is evaluated on $(x, y) = \gamma(t)$).

Proof. Set $F = (0, x)$. By Green's,

$$\int_{\partial D} F \cdot \mathrm{d}s = \int_D \mathrm{d}A = \mathrm{area}(D).$$

On the other hand, the left-hand side is readily seen to be equal to $\int_{\partial D} x \, \mathrm{d}y$. Similar proofs with $F = (y, 0)$, $F = (-y, x)$ yield the other results. ■

Exercise 3.1: Compute $\mathrm{area}(D)$ via Green's, where $D = \left\{ \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 \right\}$.

Solution. The boundary is given parametrically in two parts, $x \in [-a, a]$ and $y = \pm b\sqrt{1 - x^2/a^2}$. The plus and minus of y give the same so by the previous corollary,

$$\mathrm{area}(D) = 2b \int_{-a}^{a} \sqrt{1 - x^2/a^2} \, \mathrm{d}x = \pi ab,$$

where we compute the final integral via a trig substitution. ■

↪**Corollary 3.3**: Let $D = D_1 \setminus D_2$ where $\partial D_1, \partial D_2$ are piecewise C^1 with orientations induced from $D_1 + D_2$. If $F = (P, Q)$ a C^1 vector field with $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$ on D, then

$$\int_{\partial D_1} F \cdot \mathrm{d}s = \int_{\partial D_2} F \cdot \mathrm{d}s.$$

Proof. With the given orientations, $\partial D = \partial D_1 - \partial D_2$. By Green's, $\int_{\partial D} F \cdot \mathrm{d}s = 0$ since $P_x = Q_y$. On the other hand, by linearity and given the orientations, $\int_{\partial D} F \cdot \mathrm{d}s = \int_{\partial D_1} F \cdot \mathrm{d}s - \int_{\partial D_2} F \cdot \mathrm{d}s$. Combined, these give the answer. ■

↪**Theorem 3.5**: With the same assumptions as in Green's,

$$\int_{\partial D} F \cdot n \, \mathrm{d}s = \iint_D \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \, \mathrm{d}A$$

where n the inward normal vector to ∂D.

Proof. Suppose ∂D parametrized by $(x(t), y(t))$ for $t \in [a, b]$, so that $n = \frac{-\dot{y}, \dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}}$.

$$\begin{aligned}
\int_{\partial D} F \cdot n \, \mathrm{d}s &= \int_a^b (P, Q)(x(t), y(t)) \cdot n(t) \sqrt{x'(t)^2 + y'(t)^2} \, \mathrm{d}t \\
&= \int_a^b (P(x(t), y(t)), Q(x(t), y(t))) \cdot (y'(t), x(t)) \, \mathrm{d}t \\
&= \int_a^b P(x(t), y(t)) y'(t) \, \mathrm{d}t - \int_a^b Q(x(t), y(t)) x'(t) \, \mathrm{d}t \\
&= \int_{\partial D} \tilde{F} \, \mathrm{d}s, \quad \tilde{F} := (-Q, P) \\
\text{Green's} \quad &= \iint_D \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \, \mathrm{d}A.
\end{aligned}$$

At the penultimate line, one can alternatively arrive at the conclusion by the abuse of notation

$$\int_a^b P(x(t), y(t)) y'(t) \, \mathrm{d}t = \int_{\partial D} P \, \mathrm{d}x = \iint_D \frac{\partial P}{\partial x} \, \mathrm{d}A$$

also by Green's. ■

3.5.1 Harmonic Functions

Let $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ be the Laplacian operator in two variables. We call a function $u \in C^2(\Omega)$ *harmonic* provided $\Delta u \equiv 0$ in Ω.

↪**Corollary 3.4**: Let u be a harmonic function in a domain $D \subset \mathbb{R}^2$ with piecewise C^1 domain. Then $\int_{\partial D} \nabla u \cdot n \, \mathrm{d}s = 0$.

Proof. Take $F = \nabla u$ in the previous corollary. ■

↪**Corollary 3.5** (Green's Identities): Let $\varphi, \psi \in C^2(D)$, where $D \subset \mathbb{R}^2$ with piecewise C^1 domain with positive orientation. Then the following hold:

1. $\iint_D \varphi \Delta \psi \, \mathrm{d}A = -\iint_D \nabla \varphi \cdot \nabla \psi \, \mathrm{d}A + \int_{\partial D} \varphi \nabla \psi \cdot n \, \mathrm{d}s$
2. $\iint_D [\varphi \Delta \psi - \psi \Delta \varphi] \, \mathrm{d}A = \int_{\partial D} \varphi \nabla \psi \cdot n - \psi \nabla \varphi \cdot n \, \mathrm{d}s$

Proof.

1. follows from the previous-previous corollary applied to $F = \varphi \nabla \psi$.
2. follows from 1. by swapping the roles of φ, ψ and subtracting the following identities from each other.

■

§3.6 Stoke's Theorem

↪**Definition 3.16**: Let $F = (F_1, F_2, F_3) : \mathbb{R}^3 \to \mathbb{R}^3$ a C^1-vector field. Define

$$\begin{aligned} \operatorname{curl} F &:= \det \begin{pmatrix} e_1 & e_2 & e_3 \\ \partial_x & \partial_y & \partial_z \\ F_1 & F_2 & F_3 \end{pmatrix} \\ &:= \left(\partial_{x_2} F_3 - \partial_{x_3} F_2, \partial_{x_3} F_1 - \partial_{x_1} F_3, \partial_{x_1} F_2 - \partial_{x_2} F_1 \right). \end{aligned}$$

↪**Theorem 3.6** (Stoke's): Let S be a compact, oriented surface with positive orientation and piecewise C^1 boundary ∂S. Let $F : S \to \mathbb{R}^3$ a C^1 vector field. Then

$$\iint_S \operatorname{curl} F \cdot \mathrm{d}S = \int_{\partial S} F \cdot \mathrm{d}s.$$

Proof. Recall that for a vector field G, $\int_S G \cdot \mathrm{d}S := \int_S G \cdot n \, \mathrm{d}S$ where n the (in this case, outward) unit normal to S.

Step 1: assume first S is C^2 and ∂S is piecewise C^1. Since S regular, for all $p \in \partial S$, S is locally given by a graph over a a coordinate plane by the implicit function theorem. For instance if, suppose, S is locally given by $z = g(x, y)$, then $\Phi(x, y) = (x, y, g(x, y)) \in S$ gives a local parametrization of S, and induces positive unit normal

$$n = \frac{-g_x, -g_y, 1}{\sqrt{1 + g_x^2 + g_y^2}}.$$

We can always pick g such that $(0,0,g(0,0)) = p$. By compactness, we can decompose S into a finite union of "subsurfaces"

$$S = \bigcup_{\ell=1}^{N} S_\ell.$$

such that $S_\ell^{\text{int}} \cap S_k^{\text{int}} = \varnothing$ for all $k \neq \ell$, and such that each S_ℓ is locally a graph over a coordinate plane for each $\ell = 1, \ldots, N$. Then, we note that

$$\int_{\partial S} F \cdot \mathrm{d}s = \sum_{\ell=1}^{N} \int_{\partial S_\ell} F \cdot \mathrm{d}s, \quad \iint_S \operatorname{curl} F \cdot \mathrm{d}S = \sum_{\ell=1}^{N} \iint_{S_\ell} \operatorname{curl} F \cdot \mathrm{d}S,$$

since the overlapping boundary regions will cancel each other out by orientation. Thus, it suffices to prove Stoke's in the case that S is a graph. By rotation, translating, and rescaling, we may assume that S is a C^2 graph over $D \subset \mathbb{R}^2_{x,y}$ with $D = [-1,1] \times [0,1]$. Then, on the one hand,

$$\int_S \operatorname{curl} F \cdot \mathrm{d}S = \iint_D \big(\partial_y F_3 - \partial_z F_2\big)(-g_x) + (\partial_z F_1 - \partial_x F_3)\big(-g_y\big) + \partial_x F_2 - \partial_y F_1 \, \mathrm{d}A.$$

We now consider the integral over ∂S. We see first that $\partial S = \Phi(\partial D) = \{(x,y,g(x,y)) : (x,y) \in \partial D\}$. Thus, by chain rule, $\mathrm{d}s = \big(\mathrm{d}x, \mathrm{d}y, g_x \, \mathrm{d}x + \mathrm{d}g_y \, \mathrm{d}y\big)$ so

$$\int_{\partial S} F \cdot \mathrm{d}s = \int_{\partial D} (F_1 + F_3 g_x) \, \mathrm{d}x + \int_{\partial D} \big(F_2 + F_3 g_y\big) \, \mathrm{d}y = \int_{\partial D} G \cdot \mathrm{d}s,$$

where

$$G(x,y) := \big(F_1 + F_3 g_x, F_2 + F_3 g_y\big),$$

where, crucially, each F_i is evaluated at $(x,y,g(x,y))$. Applying Green's to this integral and applying a lot of chain and product rule gives the result; one will expect higher-order derivatives of g to appear, but these will (should!) cancel.

Step 2: Suppose now $S \in C^1$ with pw-C^1 boundaries. The heuristic idea is to "approximate" S with C^2 surfaces and show that this approximation respects the identity. We won't show the details. ■

↪**Corollary 3.6**: Suppose S_1, S_2 are two oriented surfaces with $\partial S_1 = \partial S_2 = \gamma$, and suppose S_1 and S_2 induce opposite orientations on γ. Then, for any $F \in C^1$,

$$\iint_{S_1} \operatorname{curl} F \cdot \mathrm{d}S = -\iint_{S_2} \operatorname{curl} F \cdot \mathrm{d}S.$$

↪**Corollary 3.7**: With S as above, suppose $\partial S = \varnothing$ and $F \in C^1(S)$. Then $\iint_S \operatorname{curl} F \cdot \mathrm{d}S = 0$.

↪**Corollary 3.8**: Suppose $F = \nabla f$ for some $f \in C^2$ and S is an oriented surface with $\partial S = \gamma_1 - \gamma_2$. Then

$$\int_{\gamma_1} F \cdot \mathrm{d}s = \int_{\gamma_2} F \cdot \mathrm{d}s.$$

Proof. One checks that $\operatorname{curl}(\nabla f) = 0$. ■

§3.7 Divergence Theorem

↪**Theorem 3.7**: Let $W \subset \mathbb{R}^3$ be a bounded, solid region in $\mathbb{R}^3$ with a piecewise smooth boundary $S = \partial W$ with positive orientation, and F a C^1 vector field on $W \to \mathbb{R}^3$. Then,

$$\iiint_W \operatorname{div} F \, \mathrm{d}V = \iint_S F \cdot \mathrm{d}S,$$

where if $F = (F_1, F_2, F_3)$, $\operatorname{div}(F) = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$.

↪**Corollary 3.9**: Under the same assumptions as above, if $f, g \in C^2(W)$, then

$$\iiint_W [g\Delta f - f\Delta g] \, \mathrm{d}V = \iint_{\partial W} [g\nabla f \cdot \mathrm{d}S - f\nabla g \cdot \mathrm{d}S].$$

Proof. As in Stoke's theorem, it suffices to prove for W a simple region. Clearly, it suffices to prove that

$$\iiint_W \frac{\partial F_3}{\partial z} \, \mathrm{d}V = \iint_S F_3 \hat{k} \cdot \mathrm{d}S,$$

where $\hat{k} = (0, 0, 1)$, by linearity. Assume

$$W = \{(x, y, z) : (x, y) \in D, g(x, y) \leq z \leq h(x, y)\}.$$

Fubini's gives

$$\begin{aligned}\iiint_W \frac{\partial F_3}{\partial z} \, \mathrm{d}V &= \iint_D \int_{g(x,y)}^{h(x,y)} \frac{\partial F_3}{\partial z} \, \mathrm{d}z \, \mathrm{d}x \, \mathrm{d}y \\ &= \iint_D [F_3(x, y, h(x, y)) - F_3(x, y, g(x, y))] \, \mathrm{d}x \, \mathrm{d}y.\end{aligned}$$

We can decompose the boundary $\partial W = S$ into three parts,

$$\begin{aligned}S_1 &= \{(x, y, g(x, y)) : (x, y) \in D\} \Rightarrow n = -\frac{-g_x, -g_y, 1}{\sqrt{1 + g_x^2 + g_y^2}} \\ S_2 &= \{(x, y, h(x, y)) : (x, y) \in D\} \Rightarrow n = \frac{-h_x, -h_y, 1}{\sqrt{1 + h_x^2 + h_y^2}} \\ S_3 &= \{\langle n, k \rangle = 0\}.\end{aligned}$$

The relevant integral over S_3 is zero. Over S_1, we get

$$\iint_{S_1} F_3(x, y, h(x, y)) - F_3(x, y, g(x, y)) \, \mathrm{d}x \, \mathrm{d}y,$$

since $\hat{k} \cdot n = 1$ in each case. ■

↪**Corollary 3.10**: Suppose $W, W_1, \ldots, W_k \subset \mathbb{R}^n$ are regions as in the previous theorem, with

$$W_i \cap W_j = \varnothing, i \neq j, \quad W_j \subset W^{\text{int}}, \quad p_j \in W_j^{\text{int}} \, \forall \, j = 1, \ldots, k$$

for some points $p_1, \ldots, p_k$. If F a C^1 vector field on $W - \{p_1, \ldots, p_k\}$ and $\operatorname{div} F = 0$ for all $x \in W - \{p_1, \ldots, p_k\}$. Then,

$$\iint_{\partial W} F \cdot \mathrm{d}S = \sum_{j=1}^{k} \iint_{\partial W_j} F \cdot \mathrm{d}S.$$

Proof. Apply the divergence theorem to $W - \bigcup_{j=1}^{k} W_j$. ■

Exercise 3.2 (Gauss's Law): Let $r = (x, y, z)$. Then

$$\iint_{\partial W} \frac{r}{\|r\|^3} \, \mathrm{d}S = \begin{cases} 4\pi & 0 \in W \\ 0 & \text{else} \end{cases}$$

3.7.1 Application to PDEs

Let $f \in C^2(\mathbb{R}^3)$ with f compactly supported outside of $[-R, R]^3$. The problem we study is to solve

$$\Delta u = f.$$

↪**Theorem 3.8**: For $x, y \in \mathbb{R}^3$, set $K(x, y) := -\frac{1}{4\pi |x-y|}$. Then, $K(x, y)$ is called a "fundamental solution" of Δ in $\mathbb{R}^3$ and satisfies:

1. $\Delta_x K = 0$ in $\mathbb{R}^3 - \{0\}$
2. given $f \in C^2(\mathbb{R}^3)$ with f compactly supported in $[-R, R]^3$, the function

$$u(x) = \int_{\mathbb{R}^3} K(x, y) f(y) \, \mathrm{d}y, \quad x \in \mathbb{R}^3$$

is a solution to $\Delta u(x) = f(x)$.

Proof.

$$\begin{aligned} u(x) = \int_{\mathbb{R}^3} K(x, y) f(y) \, \mathrm{d}y &= \int_{\mathbb{R}^3} -\frac{1}{4\pi |x - y|} f(y) \, \mathrm{d}y \\ &= \int_{\mathbb{R}^3} -\frac{1}{4\pi |\xi|} f(x - \xi) \, \mathrm{d}x \end{aligned}$$

Since $f(x - \xi) = 0$ for $|x - \xi| \geq R$, it follows that $f(x - \xi) = 0$ if $|x| < \frac{M}{2}$ when $|\xi| > M$ for $M \gg 1$ large. Assume $|x| < \frac{M}{2}$ with $M \gg R$, so in our integral we only need to integrate over $|\xi| \leq M$:

$$u(x) = \int_{B_M(0)} \left(-\frac{1}{4\pi |\xi|} \right) f(x - \xi) \, \mathrm{d}V(\xi), \quad \forall \, |x| < \frac{M}{2}.$$

Since $f \in C^2$, we may differentiate under the integral sign, twice. This implies (adding up these partials)

$$\begin{aligned}\Delta u(x) &= \int_{B_M(0)} \frac{-1}{4\pi\,|\xi|}\Delta_x f(x-\xi)\,\mathrm{d}V(\xi)\\ &= \int_{B_M(0)} \frac{-1}{4\pi\,|\xi|}\Delta_\xi f(x-\xi)\,\mathrm{d}V(\xi) \quad \text{(chain rule)}.\end{aligned}$$

We aim to apply Green's identity, but need to deal with the singularity at $\xi = 0$. Let $\delta > 0$ small. We rewrite the integral as

$$\Delta u(x) = \int_{B_\delta(0)} (\ldots) + \int_{W_\delta} (\ldots) =: I_1 + I_2, \quad W_\delta := B_M(0) \setminus B_\delta(0).$$

We show that $I_1 \to 0$ as $\delta \to 0$. By change of variables to spherical coordinates and the fact that $\Delta_\xi f(x-\xi)$ bounded on compact sets,

$$\begin{aligned}|I_1| &\le \int_{B_\delta} \frac{C}{|\xi|}\,\mathrm{d}\xi\\ &\le C\int_0^\delta \int_0^{2\pi} \int_0^\pi \frac{1}{\rho}\rho^2 \sin(\varphi)\,\mathrm{d}\varphi\,\mathrm{d}\theta\,\mathrm{d}\rho \quad \text{(by change of variables, Fubini)}\\ &= C'\delta^2 \underset{\delta\to 0}{\to} 0.\end{aligned}$$

Next we deal with I_2. Note that $\frac{1}{|\xi|}$ is harmonic on W_δ (i.e. away from the singularity at 0). Thus, by Green's identity (and the fact that f vanishes identically on $\partial B_M(0)$),

$$\begin{aligned}I_2 &= \int_{W_\delta}\left(-\frac{1}{4\pi\,|\xi|}\right)\Delta_\xi f(x-\xi)\,\mathrm{d}V(\xi) + \underbrace{\int_{W_\delta} \Delta_\xi\left(\frac{1}{4\pi\,|\xi|}\right) f(x-\xi)\,\mathrm{d}V(\xi)}_{=0}\\ \text{(Green's)} \quad &= \int_{\partial B_\delta(0)} f(x-\xi)\nabla_\xi\left(-\frac{1}{4\pi\,|\xi|}\right)\cdot \mathrm{d}S(\xi) - \int_{\partial B_\delta(0)}\left(-\frac{1}{4\pi\,|\xi|}\right)\nabla_\xi f(x-\xi)\cdot \mathrm{d}S(\xi)\\ &=: I_3 + I_4.\end{aligned}$$

We claim $|I_4| \to 0$ as $\delta \to 0$. Note that $|\xi| \equiv \delta$ on $\partial B_\delta(0)$ and $\nabla_\xi f(x-\xi)$ is uniformly bounded on compact sets. Thus,

$$|I_4| \le \frac{C}{\delta}\left|\int_{\partial B_{\delta(0)}} \mathrm{d}S\right| = \frac{C'}{\delta}\cdot\delta^2 \overset{\delta\to 0}{\to} 0,$$

using that that surface area of a sphere in $\mathbb{R}^3$ is proportional to its radius squared.

We finally just have to deal with I_3. We compute, expanding the gradient, that

$$\begin{aligned}I_3 &= \int_{\partial B_\delta(0)} f(x-\xi)\left(\frac{\xi}{4\pi\,|\xi|^3}\right)\cdot \mathrm{d}S_\xi\\ &= \frac{1}{4\pi\delta^3}\int_{\partial B_\delta(0)} f(x-\xi)\xi\cdot\frac{\xi}{\delta}\,\mathrm{d}S_\xi\\ &= \frac{1}{4\pi\delta^2}\int_{\partial B_\delta(0)} f(x-\xi)\,\mathrm{d}S_\xi.\end{aligned}$$

Since $f \in C^1$, for δ sufficiently small, $|\xi| < \delta$ implies $f(x-\xi) = f(x) + E(x,\xi)$ where $E(x,\xi)$ goes to zero like δ as $\delta \to 0$, by Taylor expansion. Thus,

$$
\begin{aligned}
I_3 &= \frac{1}{4\pi\delta^2}\int_{\partial B_\delta(0)} f(x) + E(x,\xi)\,\mathrm{d}S_\xi \\
&= f(x) + \frac{1}{4\pi\delta_2}\underbrace{\int_{\partial B_\delta(0)} E(x,\xi)\,\mathrm{d}S_\xi}_{\leq C\delta^2\cdot\sup_\xi|E(x,\xi)|\leq C'\delta^3},
\end{aligned}
$$

where we use the fact that $\int_{\partial B_\delta(0)} \mathrm{d}S = 4\pi\delta^2$. Based on the bound on the integral involving $E(x,\xi)$, the remaining term goes to zero as $\delta \to 0$. In short, we have that $I_3 \to f(x)$ as $\delta \to 0$. Combining all of this work gives the result, $\Delta u(x) = f(x)$ for all $|x| < \frac{M}{2}$. Since M arbitrary, the result is proven. ■

§4 Practice Questions

Question 4.1: *Let $f(x,y) = \frac{x^4y^2}{x^2+y^2}$ when $(x,y) \neq 0$ and $f(0,0) = 0$. Prove that f is differentiable everywhere in $\mathbb{R}^2$.*

Solution. If $(x,y) \neq (0,0)$, one can take partial derivatives and find that they are both continuous, thus f differentiable away from $(0,0)$. At $(0,0)$, we claim $Df(0,0) = 0$. For $h = (h_1, h_2) \in \mathbb{R}^2$, we find

$$\frac{|f(h_1,h_2) - f(0,0)|}{\|h\|} = \underbrace{\frac{h_1^4}{\left(h_1^2+h_2^2\right)^{1/2}}}_{\leq h_1^3} \underbrace{\frac{h_2^2}{h_1^2+h_2^2}}_{\leq 1} \leq h_1^3 \underset{\|h\|\to 0}{\to} 0.$$

■

Question 4.2: *Given the curve $C = \{y^2 + xy + \sin x = 1\}$, prove that C can be written as $y = g(x)$ for some C^1 function g near $\left(\frac{\pi}{2}, 0\right)$. Is the same true near $(0,1)$?*

Solution. Let $G(x,y) = y^2 + xy + \sin x - 1$. This is clearly C^1. We compute

$$\nabla G(x,y) = (y + \cos x, 2y + x).$$

At $\left(\frac{\pi}{2}, 0\right)$, we see $\nabla G\left(\frac{\pi}{2}, 0\right) = \left(0, \frac{\pi}{2}\right)$ ■

Question 4.3: *Set $f(x,y) = \frac{\sin(xy)}{xy}$ when $xy \neq 0$ and $f(0,y) = f(x,0) = 0$. Prove that f is Riemann integrable on $B := [-1,1] \times [-1,1]$.*

Solution. Its clear that $f(x,y)$ is continuous away from the line segments $L_x = \{(x,0) : -1 \leq x \leq 1\}, L_y = \{(0,y) : -1 \leq y \leq 1\} \subset B$. Thus, if we can show content$\left(L_x \cup L_y\right) = 0$, we'll be done. Let $\varepsilon > 0$. There exists some N such that $[-1,1]$ is covered by N disjoint open intervals (x_i, x_{i+1}) of length ε, by compactness. Let $B_i = (x_i, x_{i+1}) \times \left(-\frac{1}{2N}, \frac{1}{2N}\right)$ be corresponding open boxes. Its clear then that $L_x \subset \bigcup_{i=1}^N B_i$, and moreover $\mathrm{vol}(B_i) = \frac{\varepsilon}{N}$ and thus $\sum_{i=1}^N \mathrm{vol}(B_i) = \varepsilon$. This implies L_x has content zero. It follows identically that L_y has content zero, which completes the proof. ■

Question 4.4: *Find the absolute minimum and maximum values of $f(x_1, \ldots, x_n) = x_1^3 + \cdots + x_n^3$ in the closed unit ball $\overline{B_1(0)} \subset \mathbb{R}^n$.*

Solution. *(First Solution)* Notice by triangle inequality $|f(x)| \leq \sum_{i=1}^n x_i^2 |x_i| \leq 1$ over $\overline{B_1(0)}$ so $-1 \leq f(x) \leq 1$, with $f(-1,0,\ldots,0) = -1$ and $f(1,0,\ldots,0) = 1$ which means the absolute min, max of f over $\overline{B_1(0)}$ are $-1, 1$ respectively. ■

Solution. *(Second Solution)* We use Lagrange multipliers. Let $g(x) = x_1^2 + \cdots + x_n^2 - 1$. We compute $\nabla g(x) = 2x$ and $\nabla f = 3\left(x_1^2, \ldots, x_n^2\right)$. Setting $\nabla f(x) = \lambda \nabla g(x)$, we see that $2x_i = 3\lambda x_i^2$ for all i. Let $M = \{1 \leq i \leq n : x_i \neq 0\}$. Note that not all x_i's can be zero so $\#M \geq 1$, which means, for all $x_i \neq 0$, we find $x_i = \frac{2}{3\lambda}$. This implies

$$1 = \sum_{i=1}^{n} x_i^2 = \frac{4}{9\lambda^2} \cdot \#M \Rightarrow \lambda = \pm \frac{1}{\sqrt{\#M}} \frac{3}{2}$$

which implies

$$f(x) = \sum_{i=1}^{n} x_i^3 = \sum_{i \in M} \frac{8}{27\lambda^3} = \pm \frac{\#M}{(\#M)^{3/2}} \frac{8}{27} \frac{27}{8} = \pm \frac{1}{\sqrt{\#M}}.$$

It's clear that this is maximized by minimizing the size of M (which must have size at least 1) and taking the positive root (choosing λ positive), and minimized by minimizing the size of M as well, but taking the negative root. Thus if f maximized, minimized on $\partial B_1(0)$, it follows that $f(x) = 1, -1$ respectively. On the other hand, if $x \in B_1(0)$, $0 = \nabla f(x) = 3\left(x_1^2, \ldots, x_n^2\right)$ implies $x_i = 0$ for all i and thus $f(x) = 0$. Thus the min, max of f over $\overline{B_1(0)}$ are achieved on the boundary, i.e. $\min f = -1, \max f = 1$. ■

Question 4.5: *Suppose $f \in C^2(\mathbb{R}^n)$ satisfies*

$$f(x) = 2 \; \forall \; \|x\| = 1, \quad \|\nabla f(x)\| = 1, \frac{\partial^2 f}{\partial x_1^2}(x) + \cdots + \frac{\partial^2 f}{\partial x_n^2}(x) > 0 \; \forall \; x \in B_1(0).$$

Prove that $1 \leq f(x) \leq 2$ for all $x \in \overline{B_1(0)}$.

Solution. Let $x \in \overline{B_1(0)}$ and $y \in \partial B_1(0)$ the closest point (if multiple exist, choose any) to x on the boundary; in particular $\|x - y\| \leq 1$. By mean-value theorem, there exists some $z \in B_1(0)$ such that

$$f(x) - 2 = f(x) - f(y) = \nabla f(z) \cdot (x - y) \geq -\|\nabla f(z)\|\|x - y\| \geq -1 \Rightarrow f(x) \geq 1.$$

This proves the first inequality. For the second, we know the inequality is satisfied on the boundary. Suppose it fails inside $B_1(0)$. This implies, by compactness and continuity of f, that $\max_{\overline{B_1(0)}} f(x) > 2$ and moreover this max is achieved at some $x \in B_1(0)$. Being a maximum, this implies $\nabla^2 f(x)$ negative semidefinite, so in particular its eigenvalues are negative hence

$$0 \geq \sum \text{eigenvalues of } \nabla^2 f(x) = \operatorname{tr}\left(\nabla^2 f(x)\right) = \frac{\partial^2 f}{\partial x_1^2}(x) + \cdots + \frac{\partial^2 f}{\partial x_n^2}(x).$$

But by assumption, the quantity on the right is strictly positive, which is a contradiction. Thus f cannot attain its maximum inside $B_1(0)$ and thus $f(x) \leq 2$ everywhere. This completes the proof. ■